MILITARY SPECIFICATION

PROPELLENT POWDERS; GENERAL SPECIFICATION FOR METHODS OF INSPECTION, SAMPLING, AND TESTING

PART I - GENERAL INFORMATION

Section 1 - SCOPE, CONTENTS, AND NUMBERING SYSTEM

1.1 Scope. - This specification prescribes the general chemical, analytical, and physical test methods to be used in determining compliance of propellent powders with the material requirements of the applicable powder specification (see Section 2-General Provisions), and the procedures to be used in connection with the sampling and inspection of the powder.

1.2 Contents. - The contents of this specification are as follows:

Part I - GENERAL INFORMATION.
- Section 1 - Scope, Contents, and Numbering System.
- Section 2 - General Provisions.
- Section 3 - Methods of Sampling and Inspection.
- Section 4 - Notes.

Part II - TEST METHODS.
- Section 5 - Numerical Index of Test Methods.
  - Group 100 - Chemical Tests.
  - Group 200 - Analytical Tests for Organic Constituents.
  - Group 300 - Analytical Tests for Inorganic Constituents.
  - Group 400 - Stability Tests.
  - Group 500 - Physical Tests.
- Section 6 - Alphabetical Index of Test Methods.

1.3 Numbering system. - The test methods are grouped according to the property or constituent to be determined. Each method is assigned a number within its corresponding group. Each property or constitutent test method is assigned a basic number consisting of a whole number followed by a tenth decimal, while the various methods which may be used to determine the same property or constituent are differentiated from each other by consecutive designation in the tenths' decimal place following the whole number. Revision of any method for the purpose of increasing the reproducibility of the test results, or for giving additional pertinent test details will be indicated by the addition of another tenth decimal following the first decimal. Revision of a method which might significantly change the accuracy or end result of the test will be considered as a diffenent method for the property or constituent concerned, and shall be assigned a different designation in the first tenth decimal place of the basic number.

1.3.1 For example there are 4 test methods specified for the determination of diphenylamine. They are designated with the basic numbers 201.1, 201.2,

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201.3, and 201.4, respectively. A revision of method 201.1 for the purpose of increasing its reproducibility, or to give additional test details, would be designated 201.1.1; a second revision of the method would be designated 201.1.2, etc. If the revision of the method significantly changes its accuracy or the end results, the revised method would be designated 201.5.

Section 2 - GENERAL PROVISIONS

2.1 The test methods contained in this specification shall be official for use by those facilities engaged in making acceptance tests of propellent smokeless powder for loading ammunition. The tests shall be performed at a Government laboratory unless otherwise specified in the procurement documents. However, in case of conflict between the methods herein specified and those specified by the applicable powder specification, the latter shall have precedence.

2.2 The use of modifications of any test method contained herein must have prior approval of the Office Chief of Ordnance.

2.3 With the exception of certain special reagents specified under the individual test procedures, the reagent chemicals used shall conform to the specified Federal or Military specifications, or if these are not applicable, to the standards of the American Chemical Society.

Section 3 - METHODS OF SAMPLING AND INSPECTION

3.1 Sampling. - The Government inspector shall take the required test samples, as specified hereinafter, from each lot of propellent powder after the powder has been packed and sealed.

3.1.1 Extreme cleanliness shall be exercised in handling the chemical and stability samples of the powder. Touching them with damp or soiled hands shall be avoided.

3.1.2 Small arms powder.

3.1.2.1 Chemical, analytical, and physical tests' sample. - Six boxes shall be selected from each lot of powder used in loading small arms ammunition. The boxes shall be selected so as to be representative of the lot. An 8-ounce sample shall be taken from each selected box and placed in a rubber-stoppered glass bottle, or other approved container, sealed, and labeled to show powder designation, lot number, date loaded, pounds of powder in the lot, manufacturer's name and plant, and the powder box number. Under time limits of emergency conditions the ballistic sample of the powder (see 3.1.2.3) may be used for the chemical, analytical, and physical tests in lieu of the sample above specified.

3.1.2.2 When the samples arrive at the laboratory the containers shall be kept sealed, and shall be placed in a safe location for storage at ambient temperature. When ready for testing the containers shall be opened, and approximately 1 ounce of powder shall be transferred from each container to a container having a tight rubber stopper. The sample so obtained shall be blended well, and the resulting sample shall be used for the various chemical, analytical, and physical tests.

3.1.2.2 Ballistic sample. - The ballistic samples shall be taken at the same time that the chemical, analytical, and physical test samples are taken. For loading caliber .45 and caliber .30 carbine ammunition, a 4-ounce sample shall be taken from each of the 6 boxes selected from the lot (see 3.1.2.1).

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3.1.2.3 Surveillance test sample. - The surveillance test samples shall be taken at the same time that the other samples are taken. For the 82.2°C and 65.5°C surveillance tests a 5-pound sample shall be taken consisting of equal portions from each of the 6 selected boxes (see 3.1.2.1). The surveillance test samples shall be forwarded to the Commanding Officer, Picatinny Arsenal, Dover, N. J., in the case of Army purchases, or to the Supply Officer, Naval Powder Factory, Indian Head, Maryland, in the case of Navy purchases.

3.1.3 Cannon powder.

3.1.3.1 Ballistic sample. - The amount of powder required to make up the ballistic sample shall be taken from 7 boxes in a lot of cannon powder. The boxes shall be selected to be representative of the lot. The weight of the ballistic sample shall be as specified in the Ballistic Appendix to the powder specification. If the amount of powder required for the ballistic sample is less than 7 full boxes of the powder a portion equal to one-seventh of the required weight of the ballistic sample shall be removed from each of the 7 boxes, each portion being placed in an individual airtight container. The ballistic sample may be taken as soon after packing as desired, except that when moisture has been added to the powder in the blending tower the sample shall be taken no sooner than 48 hours after time of packing.

3.1.3.2 Chemical and stability samples. - From each of the 7 boxes selected for the ballistic samples (see 3.1.3.1) remove one-seventh of the amount of powder required for the chemical and stability samples by the applicable Ballistic Appendix. Part of each of these portions shall be set aside for the 134.5°C and the 120°C heat tests. The remainder of the partions shall be blended thoroughly blended to form a composite chemical tests' sample.

3.1.3.3 Surveillance test sample. - Five pounds of powder shall be taken from the composite chemical tests' sample (see 3.1.3.2) for the 65.5°C surveillance test, and forwarded to the Commanding Officer, Picatinny Arsenal, Dover, N. J., in the case of Army purchases, or to the Supply Officer, Naval Powder Factory, Indian Head, Maryland, in the case of Navy Department purchases.

3.1.3.4 Packing and marking of the samples. - All test samples shall be packed in airtight containers (preferably in glass fruit jars if the samples are small, or in powder boxes which have been tested immediately prior to use and determined to be airtight). Each sample container shall be marked to show the powder designation, lot number, manufacturer's name and plant, contract number, pounds of powder in the lot, and the number of the box from which the sample was taken.

3.1.4 General directions and precautions.

3.1.4.1 All required chemical, analytical, and physical tests on any lot of powder, including the transfer of test data to the description sheets, shall be completed within 8 days from time of delivery of the sample to the testing laboratory.

3.1.4.2 Samples delivered to the laboratory in broken containers, or in containers without stoppers, or that are not correctly labeled, or that are otherwise damaged shall be discarded, and the Government inspector at the plant or other proper official notified.

3.1.4.3 All sample containers shall remain the property of the contractor and shall be returned thereto empty and clean after acceptance of the powder.

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Section 4 - NOTES

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government Procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded, by implication or otherwise, as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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PART II - TEST METHODS

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102.2	Total Moisture (Contorl Check Method)
103.1	Volatiles (Dish and Cone Method)
103.2	Volatile Solvents
103.3	Total Volatiles (Solution-Evacuation Method)
103.4	Residual Solvent
104.1	Ether Extractive Matter
105.1	Carbon Tetrachloride Extractive Matter
106.1	Ash (No Inorganic Constituents)
106.2	Ash (Potassium Sulfate or Nitrate Present)
106.3	Ash (Potassium Sulfate and Tin Present)

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204.1	Dibutylphthalate (Difference Method)
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205.1	Dinitrotoluene (Ester Nitrates Absent)
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206.1	Trinitrotoluene (Titration Method)
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GROUP 300 - ANALYTICAL TESTS FOR INORGANIC CONSTITUENTS

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GROUP 100

CHEMICAL TESTS

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Method No. 101.1

EXTERNAL MOISTURE (SINGLE BASE POWDER)

1. PROCEDURE

1.1 Transfer an accurately weighed portion of about 10 grams of powder to a tared aluminum dish or glass weighing bottle about 60 mm in diameter and 30 mm deep. Heat the dish and contents for six hours in a vacuum oven maintained at a pressure of 80 ± 10 mm of mercury and 55° ± 2°C, cool in a desiccator and weigh. Calculate the loss in weight to percentage of external moisture in the sample.

Method No. 101.2

EXTERNAL MOISTURE (DOUBLE BASE POWDER)

1. PROCEDURE

1.1 Transfer an accurately weighed portion of five grams of the powder to a tared Erlenmeyer flask having a ground glass stopper. The flask shall have the following dimensions:

Base	40 mm
Height	80 mm
Diameter of neck opening	15 mm
Capacity	30 ml

Heat the flask and contents, with the stopper removed, in an oven maintained at 100° ± 2°C for 2 hours, at atmospheric pressure. At the end of this period, place the stopper in the flask, cool the flask and contents in a desiccator for one-half hour and weigh. Calculate the loss in weight to percentage of external moisture and volatiles in the sample.

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Method No. 102.1

TOTAL MOISTURE

1. PROCEDURE

1.1 Transfer a weighed portion of 100 to 200 g of the sample to a 500 ml balloon flask and add 200 ml of carbon tetrachloride. Fill the graduated portion of a moisture tube (See Figure 1) with carbon tetrachloride and attach the tube to the flask and condenser of the Allihn type. Connect a calcium chloride tube to the top of the condenser to keep out atmospheric moisture. Heat the flask by means of a suitable steam or hot water bath so that the distillate falls from the end of the condenser at the rate of two to three drops per second. Continue the distillation until the water layer in the measuring tube remains constant for three successive readings taken at 30-minute intervals. Disconnect the apparatus and wash the water layer with fresh carbon tetrachloride from a burette at the rate of approximately 5 ml per minute, until the volume of the water layer becomes constant. This usually requires 25 to 50 ml of carbon tetrachloride. Read the point of contact of the top meniscus with the wall of the tube and the highest point of the lower meniscus. Determine the difference in these readings and record this as the volume of water found.

2. CALCULATION

2.1 Percentage of total moisture $= \frac{100\ V}{W}$

where

V = volume of water
W = weight of sample

2.2 In the event that the volume of water distilled over is 0.2 ml or less, a satisfactory reading cannot be obtained. In this case, add, by means of an accurate pipette, 1.00 ml of water to the graduated portion of the measuring tube, then read, and correct for the volume of water added.

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Method No. 102.2

TOTAL MOISTURE (CONTROL CHECK METHOD)

1. PROCEDURE

1.1 Place two accurately weighed portions of about five grams each of the powder on suitable watchglasses. Heat these samples for one hour in an oven maintained at atmospheric pressure and a temperature of 100° ± 2°C. At the end of this period cool the samples in a desiccator and weigh. Calculate the loss in weight to percentage of moisture in the sample.

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Method No. 103.1

VOLATILES (DISH AND CONE METHOD)

1. PROCEDURE

1.1. Weigh accurately a 10 gm portion of the sample cut into small pieces and place in a tared aluminum dish with a glass cone-shaped cover. (See fig. 2). Place the apparatus on a closed steam bath and heat for 2 hours. Cool the apparatus in a desiccator containing calcium chloride and weigh. Calculate the loss in weight as percent volatiles.

Method No. 103.2

VOLATILE SOLVENTS

1. PROCEDURE

1.1 Calculate the percentage of volatile solvents by subtracting from 100 percent the sum of the percentages of external moisture, ether extractive matter, inorganic salts, and nitrocellulose in the sample as received.

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Method No. 103.3

TOTAL VOLATILES (SOLUTION EVACUATION METHOD)

1. PREPARATION FOR TEST

1.1 Dibutylphthalate reagent.—Prepare a large quantity of dibutylphthalate reagent containing approximately 1 g of diphenylamine per liter as follows: Dissolve 1 g of diphenylamine in 100 ml of hot dibutylphthalate. Pour this solution into 500 ml of dibutylphthalate and shake vigorously. Add dibutylphthalate to make 1 liter. Shake vigorously. (To make more or less reagent change the amount of the chemical proportionately.) When tested separately in the solution-evacuation apparatus 50 ml of dibutylphthalate reagent should lose less than 10 mg. It may be necessary to pretreat the dibutylphthalate as follows to meet this requirement. Heat the dibutylphthalate for 2 hours at 145° to 150°C while bubbling dry air through it.

1.2 Preparation of solution tubes.—Place 10 clean dry steel balls, 5/16 inch in diameter, and 50 ml of dibutylphthalate reagent into each of a series of clean dry solution tubes (see fig. 3). The dibutylphthalate reagent may be added through a small funnel either by means of a graduated cylinder or from an automatic 50 ml delivery device provided with a larger-than-normal outlet. Clean the inlet of each solution tube with absorbent cotton held by steel forceps. Using the forceps, insert a wad of pyrex glass wool, or equivalent, (about 0.2 g) in the inlet of each tube at such a point that it will not be in danger of falling into the tube and at the same time will not be in too close contact with the cork stopper that is to be subsequently inserted. By means of 5 cm pieces of rubber tubing[1] attach each solution to the manifold of a rocking device inside an oven[2] maintained at 85° ± 1°C. The oven should be fitted with a rocking device and electric motor. By means of the rocking device the tubes are tilted back and forth through an angle of approximately 90° at the rate of 12 complete cycles per minute.

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1 Rubber pressure tubing should have a 5/16 inch bore, 3/16 inch wall. Use 5 cm pieces for connecting the glass tubes to the copper manifold. Cut off a number of pieces, boil these in 5 percent sodium hydroxide solution for 10 to 15 minutes, wash thoroughly, and dry at 100°C. The rubber tubing should be examined after each volatile solvent determination and replaced when necessary.

2 The oven designated as Catalog No. 95105-A by the Central Scientific Co. has been found satisfactory and should be provided with a safety latch to replace the regular latch. A steam oven, if available, may be used in place of an electric oven.

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Attach to the manifold a vacuum line assembly[3] (see fig. 6) and a vacuum pump which is capable of consistently maintaining a pressure of 1 mm or less. Close stopcock G. With stopcock A open and stopcock B and needle valve E closed, turn on the rocking motor and vacuum pump, and evacuate the tubes to a pressure of 1 mm or less as indicated by the McLeod gage. This pretreatment procedure is accomplished very easily because the evacuation is started when the dibutylphthalate reagent is still at approximately room temperature. The low boiling impurities are removed gradually at 1 mm pressure as the temperature rises. At the end of 1-1/2 hours, stop the vacuum pump and rocking motor, open stopcock B cautiously, and admit dry air slowly by gradually opening the needle valve E. If air is admitted too rapidly, the wads of pyrex wool, or equivalent may be drawn into the solution tubes. Wearing a pair of gloves, remove the solution tubes from the oven, leaving the 5 cm rubber connecting tubes attached, and place the solution tubes in a wire-screen tray (see fig. 5). Immediately insert small protective drying tubes[4] (see fig. 8), and allow the tubes and contents to cool to room temperature. This will require from 30 to 45 minutes. The small drying tubes will prevent the admission of atmospheric moisture during the cooling period.

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3 Vacuum line assembly should be as shown on figure 6. The air and vapors from the solution tubes are passed through a T-tube leading to a McLeod gage, a T-tubing leading through an aspirator bottle I, filled with drierite, to a needle valve E, a three-way stopcock F, a 500 ml absorption bottle H containing silica gel, a trap D, a T-tube leading to a stopcock G, and a pressure regulator C (see fig. 6). The bottle of drierite I serves to dry the air which is admitted to the tubes in the oven. The needle valve E permits a much closer control of the pressure during the initial stage of the evacuation than can be obtained by stopcock B. The three-way stopcock F serves to admit nitrogen to the tubes in the oven. The absorption bottle of silica gel H is included to absorb the volatiles from the powder and thus make it unnecessary to change the oil in the vacuum pump frequently. The silica gel may become ineffective as shown by the inability of the vacuum pump to exhaust the system to 1 mm or less. If this happens, the silica gel should be replaced with fresh material. The trap D serves to keep dibutylphthalate reagent, in the pressure regulator, away from absorption bottle H in case stopcock A is accidently closed when the vacuum pump is stopped. The stopcock G serves to admit air into the vacuum pump and pressure regulator after the tubes have been filled with nitrogen. The regulator C, containing dibutylphthalate reagent serves to maintain the desired interior pressure.

4 Small protective drying tubes-16 are required. These tubes contain indicating drierite (or equivalent). Store them in a desiccator when not in use. Keep the drierite anhydrous (blue) by occasionally putting the tubes in an oven at 130° to 150°C overnight.

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1.3 Determination of loss of weight of solution tubes.— The determination of the loss in weight of a solution tube involves the measurement of a decrease of 20 to 40 mg (for powder containing 1 to 2 percent total volatiles) in the weight of a 170 g system. Therefore, the weighing and conditioning procedure is very important. After the tubes have cooled to room temperature, keep them stoppered, except while being weighed, to prevent absorption of atmospheric moisture by the very dry dibutylphthalate reagent. Remove the rubber connectors and protective drying tubes from all of the solution tubes and insert No. 1 cork stoppers. Wipe the tubes with a wet towel and then wipe (but do not rub) with a clean dry cloth which is substantially free from lint. This operation is for the purpose of cleaning the tubes, removing electrostatic charges, and hastening the attainment of equilibrium with moisture content of the air. Place the solution tubes in a wire-screen tray, cover them with a cloth, and allow them to stand near the balance for at least 30 minutes. At no time after a tube has been conditioned should it be touched by the fingers. In putting a tube on the balance to weigh it, handle it only by its inlet tube after covering the latter with a small piece of tissue paper. Place each solution tube in a wire holder (see fig. 9) and weigh.

2. TEST PROCEDURE

2.1 Small grains.—

2.1.1 For specimens consiting of small grains where each grain weighs less than 0.2 g, take approximately 2 g of whole grains of powder for each determination. Weigh the samples to the nearest 0.01 g in an aluminum weighing dish, place them in a series of small weighing bottles, and stopper the bottles immediately.

2.2 Large grains.—

2.2.1 For specimens consisting of large grains, where each grain weighs 0.2 to 0.4 g, take 1/2 of each of 20 to 10 grains, so that each sample will weigh approximately 2 g for each determination. These small grains may be cut with heavy tin snips inside a 1 gallon container if desired, but is is preferable to cut several grains at a time with a powder cutter[6]. If each grain weighs more than 0.4 g, cut approximately 4 g from the centers of 10 grains with the powder cutter. Immediately after cutting the slices, place them in a weighing bottle and stopper it at once. When the slicing operation has been completed, subdivide the slices with tin snips so the fragments can be easily placed in solution tubes. Divide the fragments into 2 approximately equal portions and put each one into a stoppered weighing bottle and weigh to the nearest 0.01 g.

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6 A powder cutter is a modified paper cutter with a 16 inch blade. The stationary guide plate adjacent to the blade has been provided with 16 holes ranging in size from one-eighth to three-fourths inch in which to insert the powder grains. A movable guide plate behind the stationary plate but parallel to it is actuated by a screw. The movable plate is provided with pins which fit into the holes in the stationary plate when the two plates are brought together. To aid in cutting thin uniform slices, the wheel actuating the movable plate is graduated on its circumference in small divisions. By rotating the wheel one small division to the left, the movable plate is moved forward 0.17 mm. By inserting grains in the holes of the stationary plate, turning the wheel the proper amount, and operating the cutting lever, slices of any desired thickness can be cut.

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2.3 Weigh 2 or 3 of the tubes accurately for use in making blank determinations; that is, determination of the small change in weight that occurs when the tubes of dibutylphthalate reagent are heated and evacuated under the same conditions that prevail in the analysis, making these weighings, as well as all other accurate ones, by using a counterpoise[5] (see fig. 4) on the right-hand side of the balance. Keep this counterpoise near the balance so that it will be practically in equilibrium with the prevailing conditions of temperature, pressure, and humidity. Do not wipe the counterpoise when the solution tubes are wiped. Make at least 2 blank determinations (the first and eighth tubes) if a total of 8 tubes is being used, but make at least 3 blank determinations if a total of 16 tubes is being used, (the first, eighth, and sixteenth tubes). Add the weighed samples to the tubes and weigh the tubes accurately as described in paragraph 2.3.1.

2.3.1 From each tube remove the cork stopper, remove the wad of pyrex wool with forceps, add the sample with the aid of a small metal funnel having an outlet of approximately 9 mm outside diameter, reinsert the wad of pyrex wool, or equivalent, weigh the tube accurately (see paragraph 1.3), shake it if necessary to loosen any grains which may be adhering to the walls of the tube, connect it to the manifold in the electric oven, and rock the tube mechanically. It is considered desirable, although not essential to have the samples dissolve completely during the overnight rocking period at 85°C, and they are much more likely to dissolve if the steel balls and grains of powder all move freely in the tubes when they are put in the oven. If the samples are of such a nature that they do not adhere firmly to the walls of the tubes on standing in the dibutylphthalate solutions for 30 minutes (this must be determined by experience), the procedure just described may be shortened in two respects: (1) the weighed samples may be transferred directly from the aluminum weighing dish to the solution tubes and (2) the whole series of tubes may be put into the oven at the same time. In some cases it has been found advantageous to rock the weighed solution tubes in a cradle from the time they are weighed so that all can be put into the oven at the same time.

---

5 A counterpoise tube is a glass tube containing a suitable number of steel balls so that it will have approximately the same weight, volume, and exterior surface as a solution tube containing 50 ml of dibutylphthalate. This tube should be kept standing near the balance, covered with a cloth to protect it from dust.

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2.4 As soon as all the tubes have been placed in the oven, turn on the rocking motor. Open the nitrogen cylinder valve to allow a slow stream of nitrogen to escape through the mercury trap. With stopcock A open and stopcock B closed evacuate the tubes, fill them with nitrogen, evacuate them a second time, and again fill them with nitrogen. This is accomplished very easily by manipulating stopcock F. The nitrogen should be admitted rather slowly to prevent the gas stream from carrying the glass wool or any particles of foreign matter from the tubber tubing or manifold into the tubes. Leave stopcock F turned to connect the oven with the tube leading to the nitrogen overflow bottle containing the mercury. Then admit air to the pressure regulator, vacuum pump, etc., by opening stopcock G. The operation of filling the tubes with nitrogen is facilitated by having them at approximately room temperature at this point so that they can be evacuated quickly without taking any special precautions. Rock the tubes at 85° ± 1°C for approximately 15 hours. Ordinarily the samples dissolve in the dibutylphthalate reagent under these conditions. It is essential that the samples be softened by dibutylphthalate reagent, but not essential that they dissolve completely. After the tubes have been rocked for approximately 15 hours, turn stopcock F to connect oven to vacuum pump and remove the volatile matter by evacuation at 5 mm pressure for 2 hours. The beginning of the evacuation, while most of the volatile matter is being removed, is the critical point of the determination. Assuming that the tubes are properly conditioned and weighed, the accuracy of the results depend upon the care with which this operation is performed. With the rocking device in operation and with stopcocks A and B and the needle valve E open, turn on the vacuum pump. Evacuate at gradually decreasing pressures by slowly closing needle valve E until a pressure of 5 mm is reached. The lowering of the pressure from about 50 mm to 5 mm must be done cautiously over a period of approximately 10 minutes while watching the operation through the glass door of the oven. Close stopcocks A and B and continue the evacuation for 2 hours at 85°C and 5 mm. The pressure must be reduced so gradually that bubbles of volatile matter leave the dibutylphthalate solution without danger of causing mechanical loss. At the end of 2 hours, stop the vacuum pump and rocking motor. Open stopcocks A and B, and gradually admit air into the system through needle valve E. If stopcock A is not opened at this point, the dibutylphthalate solution will flow out of the pressure regulator. Remove the tubes from the oven, insert the protective drying tubes in the rubber connectors, let the tubes cool to room temperature and insert cork stoppers. Condition and weigh the tubes in the same manner as that previously described. If a special ink has been used instead of hydrofluoric acid solution to put numbers on the tubes, care must be taken to avoid rubbing off any of the ink. As a matter of good practice and in order to obtain minimum blank values, the conditioning procedures before and after the removal of the volatile matter should be similar.

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Method No. 103.3 (contd)

$$\text{Percentage of volatile solvents} = \frac{100\ (A \neq B)}{W} - D$$

$$\text{Percentage of total volatiles} = \frac{100\ (A \neq B)}{W}$$

where

A = decrease in weight of sample tube
B = average change in weight of blank tubes
W = weight of sample
D = percentage of total moisture

The changes in weight of the blank tubes are usually negative, but may be positive (possibly due to humidity changes): they seldom exceed 3 mg. In all cases the average change in weight, B, is algebraically added to A; that is, if B is negative it is subtracted from A, if positive it is added. Blank values are usually concordant within 1 mg. If they are not concordant within 2 mg, the analysis should be repeated. This method has a precision of 0.10 percent, that is, results should agree with 0.1 unit when expressed as percent volatile solvents. If they do not agree this closely, the analysis should be repeated.

Method No. 103.4

RESIDUAL SOLVENT

1. PROCEDURE

1.1 Determine residual solvent by subtracting the amount of moisture (Method No. 102.1) from the total volatiles.

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Method No. 104.1

ETHER EXTRACTIVE MATTER

1. PROCEDURE

1.1 Cut into small pieces approximately 1/8" by 1/8" by .03" a representative portion of the sample. Weigh accurately a portion of approximately 5 grams of this prepared sample, transfer this to an extraction thimble and extract with alcohol-free anhydrous ether, using a Soxhlet or equivalent extractor. Use a hot water bath or a steam hot plate as the source of heat, and adjust the temperature so that the ether drips from the end of the condenser at the rate of 2 to 3 drops per second. Continue the extraction for at least 16 hours and then test for completeness of extraction in the following manner: Allow the extractor to become almost full of ether and then separate the extractor from the flask. Draw off the ether from the extractor into a 100 ml beaker. Transfer 10 ml of this ether to a 50 ml beaker and evaporate the ether by means of a stream of dry air. Add five drops of acetone to the beaker to dissolve any residue which remains. Test this solution for Nitroglycerin using the following test: Place 2 drops of the acetone solution, obtained as directed above, in the indenture of a white spot plate. Add about 5 mg of powdered zinc and 0.5 ml of Franchimont Reagent (prepared by adding 0.15 gm of alpha-naphthylamine and 0.50 gm of sulfanilic acid to 100 ml of 30 percent acetic acid and heating until solution is complete). The immediate appearance of a deep red color indicates the presence of Nitroglycerin and therefore incomplete extraction. Disregard the gradual appearance of a light pink coloration. If extraction is incomplete, reassemble the extraction apparatus, pour back the remaining ether which was removed from the extractor and continue the extraction until complete extraction is indicated by the test described above. When extraction is complete, evaporate the ether extract by heating on a steam bath until about 25 ml of ether remains. Complete the evaporation to dryness by means of a slow current of dry air. Retain this ether extract for subsequent determinations.

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Method No. 105.1

CARBON TETRACHLORIDE EXTRACTIVE MATTER

1. PROCEDURE

1.1 Weigh two 5 g samples into suitable extraction thimbles and extract with CCl_4 for a minimum of 16 hours. Test for completeness of extraction in the following manner: Allow the extractor to become almost full, then separate the extractor from the flask. Draw off the carbon tetrachloride into a 100 ml beaker. Transfer 10 ml of this solution into a 50 ml beaker and evaporate to dryness using a stream of dry air. Add 5 drops of acetone to dissolve any residue which remains. Test the solution for nitroglycerin using the following test: Place 2 drops of the acetone solution, obtained as directed above, in the indenture of a white spot plate. Add about 5 mg of powdered zinc and 0.5 ml of Franchimont Reagent (prepared by adding 0.15 g of alphanaphthylamine and 0.50 g of sulfanilic acid to 100 ml of 30 percent acetic acid and heating until solution is complete). The immediate appearance of a deep red color indicates the presence of nitroglycerin and therefore incomplete extraction. Disregard the gradual appearance of a light pink coloration. If extraction is incomplete, reassemble the extraction apparatus, pour back the remaining carbon tetrachloride and continue the extraction until complete as indicated by the above test. Use tared flasks and determine the percentage of carbon tetrachloride extractive matter. Determine nitroglycerin, ethyl centralite and dinitrotoluene in the extract. The residue can be used for the determination of nitroguanidine.

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Method No. 106.1

ASH (NO INORGANIC CONSTITUENTS PRESENT)

1. PROCEDURE

1.1 Transfer to a previously ignited and cooled 50-cc porcelain crucible an accurately weighed portion of approximately 2 grams of the sample. Add 5 cc of concentrated nitric acid and place the crucible on a steam bath. Allow the digestion to continue until the powder is completely disintegrated and no liquid remains. Ignite the crucible and contents by first heating gently until fumes are no longer evolved and then heating in a muffle furnace or over a gas flame until constant weight is attained. Cool the crucible and contents in a desiccator, weigh, and calculate the weight of ash as percentage in the powder as received.

Method No. 106.2

ASH (POTASSIUM SULFATE OR NITRATE PRESENT)

1. PROCEDURE

1.1 Ash a 5 gm sample as directed in Method No. 106.1, including the muffle heating. After the dish has cooled, add 10 ml of water and 5 ml of approximately 38 percent hydrochloric acid solution, then digest on the steam bath for 5 minutes. Dilute to 25 ml, add a drop of methyl orange indicator, and make slightly alkaline with approximately 5-percent ammonium hydroxide solution. Filter through an ashless filter paper, wash the dish and filter paper to remove all soluble constituents, and transfer the filter paper to the dish. Dry the dish and paper for 30 minutes at approximately 100°C, char the paper by means of a burner, ignite in the muffle at dull red heat for 15 to 30 minutes, cool in a desiccator and weigh. Calculate the increase in the weight of the dish to percentage of ash in the powder as received.

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Method No. 106.3

ASH (POTASSIUM SULFATE AND TIN PRESENT)

1. PROCEDURE

1.1 Ignite two 50-ml porcelain crucibles and lids and cool in a desiccator. Designate the pairs of crucibles and lids as assemblies A and B. Weigh assembly A using assembly B as a counterpoise. Transfer a weighed portion of approximately 5 gm of powder to assembly A and add 5 ml of concentrated HNO_3 to each crucible. Place the assemblies on a steam bath, adjust the lids to permit the escape of volatile matter, and heat until the powder has been converted to a gummy mass. Transfer the assemblies to a hot plate (approximately 250°C) and drive off volatile matter. Place in an electric muffle furnace at 600°C and ignite for approximately 2 hours to remove carbonaceous matter. Cool and add 5 drops of concentrated H_2SO_4 to each crucible. Return to the hot plate and heat till fuming ceases. Transfer to the muffle furnace, ignite for 15 minutes, remove and cool in a desiccator. Weigh assembly A using assembly B as the counterpoise. Calculate the percent ash as follows:

$$\text{Percent ash} = \frac{100A}{W} - B - 1.27C$$

where

A = increase in weight of assembly A
B = percent potassium sulfate
C = percent tin determined as specified in the detail specification
W = gm of sample

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GROUP 200

ANALYTICAL TESTS OF ORGANIC INGREDIENTS

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Method No. 201.1

DIPHENYLAMINE (TITRATION METHOD)

1. REAGENTS

1.1 Prepare the purified carbon tetrachloride by repeatedly agitating 500 ml CCl_4, free from suspended matter, with 25 ml portions of concentrated sulfuric acid until the lower (acid) layer is no longer colored, discarding the lower layer each time. Then wash the CCl_4 repeatedly with 100 ml portions of water until the aqueous (upper) layer is neutral to methyl red, discarding the upper layer each time. Transfer the CCl_4 to an all-glass distilling apparatus equipped with a fractionating column. Collect the fraction distilling at 76° to 77°C, discarding the first and last 25 ml portions.

1.2 Prepare the 0.2N potassium bromate-bromide solution by dissolving 5.568 grams of $KBrO_3$ and 30 grams of KBr in 500 ml water which contains 0.50 gram of potassium hydroxide. Dilute the solution to one liter in a volumetric flask.

1.3 Prepare the 0.2N sodium thiosulfate solution by dissolving 55 grams of $Na_2S_2O_3 \cdot 5H_2O$ and diluting to one liter, using freshly boiled and cooled water. Add one ml of chloroform and allow the solution to age for 24 hours. Standardize the solution by any acceptable analytical method.

2. DETERMINATION

2.1 Dissolve the dried ethyl ether extract (not necessary to dry to constant weight if diphenylamine content only is desired) in 10 ml of hot purified carbon tetrachloride and transfer to a 250 ml iodine flask. Rinse the original flask with four 10 ml portions of hot purified carbon tetrachloride and add the rinsings to the iodine flask. Add twenty-five ml of 0.2N potassium bromate-bromide solution, accurately measured, and 5 ml of concentrated hydrochloric acid to the flask, stopper immediately, and shake vigorously for five minutes with intermittent cooling in ice water. Place ten ml of freshly prepared 15 percent potassium iodide solution in the trough of the stoppered flask and open the stopper slightly to admit most of the solution. Wash the trough by adding 5 to 10 ml water, then allowing the water to run into the flask by again raising the stopper slightly. Shake the flask and contents for one minute, and then titrate the liberated iodine slowly with 0.2N sodium thiosulfate solution, while swirling the contents continuously, until the aqueous (upper) layer becomes pale yellow in color. Add five ml of 0.10 percent starch solution to the flask, stopper and again shake vigorously. Resume the slow addition of the thiosulfate solution, stoppering and shaking frequently. At no time shall the blue color of the upper layer be discharged while the carbon tetrachloride layer retains a purple color. Run a blank on 50 ml of the purified carbon tetrachloride contained in a 250 ml iodine flask by accurately adding 25 ml of 0.2N

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potassium bromate-bromide solution and continuing as above. Calculate the percentage of diphenylamine in the sample as follows:

$$\text{Percentage diphenylamine} = \frac{2.115\ (v-V)N}{W}$$

where

v = ml sodium thiosulfate used in blank
V = ml sodium thiosulfate used in sample
N = normality of the sodium thiosulfate solution
W = weight of sample

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Method No. 201.2

DIPHENYLAMINE (GRAVIMETRIC METHOD)

1. PROCEDURE

1.1 Dissolve the residue in the flask from the ether extractive determination (obtained as directed in Method No. 104.1) in 100 ml of ether. Transfer the ether solution to a 250-ml beaker and add 0.6 ml of bromine carefully. Place the beaker on a steam bath, maintained at approximately 75°C, and evaporate the ether and bromine with a current of dry air until a volume of approximately 10 ml is reached. Add 40 ml of approximately 95-percent (by volume) ethyl alcohol solution and heat the liquid to boiling. Add 40 ml of water at room temperature and allow the beaker to remain on the steam bath for exactly 10 minutes. Filter the hot liquid through a tared filtering crucible with a mat which will permit rapid filtration and wash the retained precipitate of tetrabromdiphenylamine with 80 ml of boiling approximately 47.5-percent (by volume) ethyl alcohol solution and finally with hot water. (The alcohol solution serves to dissolve the dinitrotoluene and dibutylphthalate, which remains in solution when the alcohol is diluted to approximately 47.5 percent by volume. This strength alcohol solution does not have appreciable solvent action on the tetrabromdiphenylamine.) Dry the crucible and contents for 1 hour at 105°C, cool in a desiccator, and weigh. Calculate the increase in weight of the crucible to percentage of diphenylamine in the powder on a moisture and volatile solvents-free basis.

$$\text{Percentage of diphenylamine} = \frac{34.87\ A}{W}$$

where

A = weight of tetrabromdiphenylamine
W = weight of sample on a moisture and volatiles free basis

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Method No. 201.3

DIPHENYLAMINE

1. PROCEDURE

1.1 Extract an accurately weighed portion of about five grams of the powder with ethyl ether (factory ether containing three to four percent alcohol) for a period of not less than 16 hours or until extraction is complete, using a Knorr, Soxhlet or other suitable extraction apparatus. The extract consists of the diphenylamine used as the stabilizer. Evaporate the extract to dryness using a current of dry air and dry the residue to constant weight in a vacuum desiccator over calcium chloride. Dissolve the residue in approximately 15 ml of factory ether and add enough bromine, drop by drop, to assure a slight excess over the amount necessary to convert all the diphenylamine to tetrabrom-diphenylamine. Evaporate the bulk of the ether using a very small current of dry air, then dry in a vacuum desiccator to constant weight. Calculate the increase in weight to percentage of diphenylamine in the sample as follows:

$$\text{Percentage of Diphenylamine} = \frac{5357\ A}{B}$$

where

A = Increase in weight of the flask and residue
B = Weight of sample

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DIPHENYLAMINE (IN PRESENCE OF CENTRALITE)

1. DESCRIPTION

1.1 In this method, centralite and diphenylamine can be determined on the same sample. The powder is digested with NaOH, and the centralite and diphenylamine are removed by steam distillation. After extraction with ether, a small amount of alcohol is added and the ether evaporated. Bromine in CCl_4 is added in slight excess and the brominated products of both Centralite and diphenylamine are filtered, dried, and weighed. Dibromcentralite is then dissolved by boiling with 68% acetic acid, in which reagent the tetrabromdiphenylamine is relatively insoluble. The latter is then filtered, dried and weighed.

2. APPARATUS

2.1 One liter balloon flask, short neck, fitted with a 2-hole rubber stopper containing (a) an inlet tube for steam, and (b) a goose neck tube about 20 cm high and 11 mm O. D. to fit into the condenser.

2.2 Two liter balloon flask, for generating steam, fitted with 2-hole stopper containing (a) exit tube for steam and (b) vertical piece of tubing about 3 feet in length as a pressure equalizer. Place a "T" tube with a short piece of rubber tubing attached, between the generator and the flask to withdraw the condensate by opening a pinch clamp on the tubing. If available, steam from a low pressure line may be used directly, using a trap to collect the condensate.

2.3 2-one liter separatory funnels.

2.4 Pyrex fritted glass crucibles, medium porosity, low form, 30 ml capacity.

2.5 18-inch Liebig condenser with adapter reaching to the liquid in the bottom of the receiving flask.

3. REAGENTS

3.1 30 percent NaOH - Dissolve 300 g NaOH in 700 ml water.

3.2 Bromine in CCl_4 - Add 20 ml of liquid bromine to 80 ml of CCl_4.

3.3 NaCl (Technical).

3.4 68 percent acetic acid. Dilute 68 ml of glacial acetic acid to 100 ml with water.

3.5 H_2SO_3, 8 percent solution, or SO_2 from a cylinder.

4. PROCEDURE

4.1 Weigh 10 g of the powder into the liter balloon flask, add 100 ml of water, 100 ml of 30 percent NaOH solution, and connect to the condenser. Place a 500 ml Erlenmeyer flask at the end of the adapter to catch the distillate. The adapter on the condenser should dip below the surface of about 25 ml of water in the flask. Connect to the steam generator and distill until a total of 350-400 ml is collected. Do not place an auxiliary heater under the flask containing the sample. With powders containing 4-5 percent centralite and 1-3 percent diphenylamine, it is necessary to collect from 450-500 ml of distillate using a 750 ml flask. Also, in this case, a 5 g sample should be used. Partially saturate the

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distillate with solid NaCl and transfer to a separatory funnel. Disconnect the condenser, clamp it in a vertical position and wash down the inside of it and adapter with 75 ml of ether, collecting the washings in the flask. Also wash the outside of the adapter and the inside of the goose neck tube with ether, catching the washings in the flask. Extract the distillate with two more 75 ml portions of ether. Transfer the combined extracts to a 400 ml beaker, add 30 ml of 95 percent ethyl alcohol and evaporate the ether on the steam bath down to a volume of about 30 ml using a current of dry air otherwise creeping of the product results. Rinse down the sides of the beaker with 20 ml of ethyl alcohol. Add dropwise from a burette the CCl_4 solution of bromine while constantly agitating the beaker until an excess is present as shown by the persistence of a reddish-light brown color. About 1.5-2.0 ml of bromine solution will be required. Crystals of dibromcentralite and tetrabromdiphenylamine precipitate after the addition of a portion of the bromine solution. After standing for 30 minutes, decolorize by passing in SO_2 or using H_2SO_3. Add 150 ml of water and heat in a steam bath with a current of air directed onto the surface of the solution until the alcohol is removed. Evaporation down to 75-100 ml is usually sufficient. After standing 10-15 minutes, filter the precipitate on a tared, fritted glass crucible and wash with water. If it is desired to determine centralite, dry the crucible and contents at 100 $\pm$ 2°C for 1 1/2 hours, cool in a desiccator and weigh. Transfer the precipitate to a 150 ml beaker. Wash the crucible with two 20 ml portions of 68 percent acetic acid, collecting the washings in the beaker. Boil the contents of the beaker for 10 minutes, then cool to room temperature and collect the undissolved tetrabromdiphenylamine in the original fritted glass crucible. Wash twice with 5 ml portions of the 68 percent acetic acid (this makes a total of 50 ml), and finally with water. Dry at 100°C for 1 1/2 hours, cool in a desiccator and weigh. Correct the weight of tetrabromdiphenylamine thus obtained for the amount dissolved by the acetic acid, by adding 0.010 g for the 50 ml used.

5. CALCULATIONS

5.1 Calculate the percentage of diphenylamine in the sample as follows:

$$\text{Percentage of Diphenylamine} = \frac{\text{Corrected wt of tetrabromdiphenylamine} \times 34.84^*}{\text{Wt of sample}}$$

$$^*\frac{169 \times 100}{485} = 34.84$$

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Method No. 202.1

ETHYL CENTRALITE (TITRATION IN TWO PHASE SYSTEM)

1. REAGENTS AND APPARATUS

1.1 0.2N $KBrO_3$ solution containing 30 gm KBr per liter.--Dry C. P. $KBrO_3$ for 2 hours at 100°C. Weigh out 5.567 gm. Dissolve in distilled water along with 30 gm of KBr and make up the solution to a liter in a volumetric flask. The solution will keep indefinitely.

1.2 1:1 HCl.--Dilute 35-37 percent HCl with an equal amount of water.

1.3 10-percent KI solution.--Dissolve 100 gm of KI in 900 gm of distilled water.

1.4 Carbon tetrachloride.--Run a blank determination on each lot of carbon tetrachloride in order to correct for errors from this source.

1.5 Flasks.-Flasks shall be stoppered iodine titration type of 250 ml capacity equipped with a gutter.

1.6 Starch solution.--To each liter of cold distilled water, add 5 gm of corn starch made into a paste with a little water. Add 10 gm of KI and 4 gm of $ZnCl_2$ crystals. Bring the mixture to boiling with agitation. Simmer for 10 minutes and continue agitating. Cool and transfer to a large bottle provided with a siphon. The starch will settle to the bottom. Use only the supernatant liquid for the determinations. The end point is sharp and the depth of the blue starch-iodine color increases with age to a very intense blue.

1.7 0.11N $Na_2S_2O_3$.--Dissolve approximately 30 gm of $Na_2S_2O_3 \cdot 5H_2O$ in a liter of distilled water. The solution should be allowed to stand for a few weeks before being standardized against the 0.2N $KBrO_3$-KBr solution. The standardization shall be conducted as follows: Add 50 ml of carbon tetrachloride to the iodine flask and add 25 ml of $KBrO_3$-KBr solution from a buret or accurate pipette. Moisten the stopper of the flask with a drop of KI solution. Add 10 ml of 1:1 HCl to the flask, stopper quickly, and shake for 5 minutes. Measure out 25 ml of 10 percent KI solution. Cool the flask under the cold water tap. Add approximately 5 ml of KI solution to the gutter of the flask, then loosen the stopper and allow the solution to run into the flask, taking as much care as possible to prevent the loss of bromine fumes. Add the remainder of the KI solution, stopper, and shake the flask for 30 seconds. Titrate the liberated iodine with the $Na_2S_2O_3$ solution. The solution finally changes from a brown to a deep violet color which on further addition of $Na_2S_2O_3$ become less intense. Near the end point, add a few ml of starch solution and continue the titration with frequent shaking until the blue color is completely discharged. Calculate the normality of the $Na_2S_2O_3$ solution as follows:

$$\text{Normality} = \frac{25 \times 0.2}{V}$$

where

V = ml of $Na_2S_2O_3$ solution used in titration.

2. PROCEDURE

2.1 Dissolve the residue in an untared flask (obtained as directed in

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Method No. 104.1) in 10 to 15 ml of carbon tetrachloride and transfer to a 100-ml volumetric flask. Rinse with several 10-ml portions of carbon tetrachloride and add to the contents of the volumetric flask. Make up to volume with carbon tetrachloride and pipette a 25-ml aliquot portion into the iodine titration flask. Add an additional 25 ml of carbon tetrachloride. Add 25 ml of the $KBrO_3$-KBr solution measured from a buret or an accurate pipette. Moisten the stopper with a drop of KI solution, add 10 ml of 1:1 HCl and quickly stopper the flask, then shake thoroughly for 1 minute. Immediately at the end of this period, add the KI and titrate the liberated iodine, following the procedure outlined in paragraph 1.7. Do not allow the mixture to stand after the initial shaking period nor after the addition of the KI, but always complete the determination without delay. Calculate the percentage of ethyl centralite as follows;

$$\text{Percent ethyl centralite} = \frac{26.82\ (V-v)\ N}{W}$$

where

V = ml $Na_2S_2O_3$ used in blank
v = ml $Na_2S_2O_3$ used in titration of sample
N = normality of $Na_2S_2O_3$
W = weight of volatiles-free sample

To insure complete bromination of the ethyl centralite at least a 100 percent excess of $KBrO_3$-KBr should be present. That is "v" should be equal to or greater than "(V-v)".

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ETHYL CENTRALITE (TITRATION IN SINGLE PHASE SYSTEM)

1. REAGENTS AND APPARATUS

1.1 Dry C.P. $KBrO_3$ for two hours at 100°C. Weigh out 5.567 g. Dissolve in distilled water along with 30 g of KBr and make up the solution to a liter in a volumetric flask. The solution will keep indefiritely.

1.2 Dilute 35-37 percent HCl with an equal amount of water.

1.3 Dissolve 100 g of KI in 900 grams of distilled water. Run a blank determination of each lot of glacial acetic acid in order to correct for errors from this source.

1.4 Flasks shall be stoppered iodine titration type of 250 ml capacity equipped with a gutter.

1.5 To each liter of cold distilled water, add 5 g of corn starch made into a paste with a little water. Add 10 g of KI and 4 g of $ZnCl_2$ crystals. Bring the mixture to boiling with agitation. Simmer for ten minutes and continue agitating. Cool and transfer to a large bottle provided with a siphon. The starch will settle to the bottom. Use only the supernatant liquid for the determinations. The end point is sharp and the depth of the blue starch-iodine color increases with age to a very intense blue.

1.6 Dissolve approximately 30 g of $Na_2S_2O_3$ $5H_2O$ in a liter of distilled water. The solution should be allowed to stand for a few weeks before being standardized against the 0.2 N $KBrO_3$-KBr solution. The standardization shall be conducted as follows: Add 50 ml of glacial acetic acid to the iodine flask and add 25 ml of $KBrO_3$-KBr solution from a burette or accurate pipette. Moisten the stopper of the flask with a drop of KI solution. Add 10 ml of 1:1 HCl to the flask, stopper quickly, and shake for five minutes. Measure out 25 ml of 10 percent KI solution. Cool the flask under the cold water tap. Add approximately 5 ml of KI solution to the gutter of the flask, then loosen the stopper and allow the solution to run into the flask, taking care to prevent the loss of bromine fumes as much as possible. Add the remainder of the KI solution, stopper, and shake the flask for 30 seconds. Titrate the liberated iodine with the $Na_2S_2O_3$ solution. The solution finally changes from a brown to a deep violet color which on further addition of $Na_2S_2O_3$ becomes less intense. Near the end point, add a few ml of starch solution and continue the titration with frequent shaking until the blue color is completely discharged. Calculate the normality of the $Na_2S_2O_3$ solution as follows:

$$\text{Normality} = \frac{25 \times 0.2}{V}$$

where V = ml of $Na_2S_2O_3$ solution used in titration.

2. PROCEDURE

2.1 Dissolve the dried residue from one of the carbon tetrachloride extraction in 10 to 15 ml of glacial acetic acid and transfer to a 100 ml volumetric flask. Rinse with several 10 ml portions of glacial acetic acid and add to the contents of the volumetric flask. Make up to

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volume and pipette a 25 ml aliquot portion into the iodine titration flask. Add an additional 25 ml of glacial acetic acid. Add 25 ml of the $KBrO_3$-KBr solution measured from a burette or an accurate pipette. Moisten the stopper with a drop of KI solution, add 10 ml of 1:1 HCL and quickly stopper the flask, then shake thoroughly for one minute. Immediately following this period, add the KI and titrate the liberated iodine, following the procedure outlined above. Do not allow the mixture to stand after the initial shaking period nor after the addition of the KI, but always complete the determination without delay. Calculate the percentage of ethyl centralite as follows:

$$\text{Percent ethyl centralite} = \frac{26.82(V-v)N}{W}$$

where

V = ml $Na_2S_2O_3$ used in blank
v = ml $Na_2S_2O_3$ used in titration of sample
N = normality of $Na_2S_2O_3$
W = weight of sample corrected for total volatiles

To insure complete bromination of the ethyl centralite at least a 100 percent excess of $KBrO_3$-KBr should be present. That is "v" should be equal to or greater than "(V-v)".

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Method No. 202.3

ETHYL CENTRALITE (IN PRESENCE OF DIPHENYLAMINE)

1. DESCRIPTION OF METHOD

1.1 In this method, centralite and diphenylamine can be determined on the same sample. The powder is digested with NaOH, and the centralite and diphenylamine are removed by steam distillation. After extraction with ether, a small amount of alcohol is added and the ether evaporated. Bromine in CCl_4 is added in slight excess and the brominated products of both centralite and diphenylamine are filtered, dried, and weighed. Dibromcentralite is then dissolved by boiling with 68% acetic acid, in which reagent the tetrabromdiphenylamine is relatively insoluble. The latter is then filtered, dried, and weighed.

2. APPARATUS

2.1 One liter balloon flask, short neck, fitted with a 2-hole rubber stopper containing (a) an inlet tube for steam, and (b) a goose neck tube about 20 cm high and 11 mm O. D. to fit into the condenser.

2.2 Two liter balloon flask, for generating steam, fitted with 2-hole stopper containing (a) exit tube for steam and (b) vertical piece of tubing about 3 feet in length as a pressure equalizer. Place a "T" tube with a short piece of rubber tubing attached, between the generator and the flask to withdraw the condensate by opening a pinch clamp on the tubing. If available, steam from a low pressure line may be used directly, using a trap to collect the condensate.

2.3 2 - one liter separatory funnels.

2.4 Pyrex fritted glass crucibles, medium porosity, low form, 30 ml capacity.

2.5 18-inch Liebig condenser with adapter reaching to the liquid in the bottom of the receiving flask.

3. REAGENTS

3.1 30 percent NaOH - Dissolve 300 g NaOH in 700 ml water.

3.2 Bromine in CCl_4. Add 20 ml of liquid bromine to 80 ml of CCl_4.

3.3 NaCl (Technical).

3.4 68 percent acetic acid. Dilute 68 ml of glacial acetic acid to 100 ml with water.

3.5 H_2SO_3, 8 percent solution, or SO_2 from a cylinder.

4. PROCEDURE

4.1 Weigh 10 g of the powder into the liter balloon flask, add 100 ml of water, 100 ml of 30 percent NaOH solution, and connect to the condenser. Place a 500 ml Erlenmeyer flask at the end of the adapter to catch the distillate. The adapter on the condenser should dip below the surface of about 25 ml of water in the flask. Connect to the steam generator and distill until a total of 350-400 ml is collected. Do not place an auxiliary heater under the flask containing the sample. With powders containing 4-5 percent centralite and 1-3 percent diphenylamine, it is necessary to collect from 450-500 ml of distillate using a 750 ml flask. Also, in this case, a 5 g sample should be used. Partially saturate the distillate with solid

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$NaCl$ and transfer to a separatory funnel. Disconnect the condenser, clamp it in a vertical position and wash down the inside of it and adapter with 75 ml of ether, collecting the washings in the flask. Also wash the outside of the adapter and the inside of the goose neck tube with ether, catching the washings in the flask. Extract the distillate with two more 75 ml portions of ether. Transfer the combined extracts to a 400 ml beaker, add 30 ml of 95 percent ethyl alcohol and evaporate the ether on the steam bath down to a volume of about 30 ml using a current of dry air otherwise creeping of the product results. Rinse down the sides of the beaker with 20 ml of ethyl alcohol. Add dropwise from a burette the CCl_4 solution of bromine while constantly agitating the beaker until an excess is present as shown by the persistence of a reddish-light brown color. About 1.5 - 2.0 ml of bromine solution will be required. Crystals of dibromcentralite and tetrabromdiphenylamine precipitate after the addition of a portion of the bromine solution. After standing for 30 minutes, decolorize by passing in SO_2 or using H_2SO_3. Add 150 ml of water and heat in a steam bath with a current of air directed onto the surface of the solution until the alcohol is removed. Evaporation down to 75 - 100 ml is usually sufficient. After standing 10-15 minutes, filter the precipitate on a tared fritted glass crucible and wash with water. Dry the crucible and contents at 100 $\pm$ 2°C for 1 1/2 hours, cool in a desiccator and weigh. The weight of the precipitate represents the combined weights of tetrabromdiphenylamine and dibromcentralite. Transfer the precipitate to a 150 ml beaker. Wash the crucible with two 20 ml portions of 68 percent acetic acid solution, collecting the washings in the beaker. Boil the contents of the beaker for 10 minutes, then cool to room temperature and collect the undissolved tetrabromdiphenylamine in the original fritted glass crucible. Wash twice with 5 ml portions of 68 percent acetic acid solution (this makes a total of 50 ml), and finally with water. Dry at 100° $\pm$ 2°C for 1 1/2 hours, cool in a desiccator and weigh. Correct the weight of tetrabromdiphenylamine thus obtained for the amount dissolved by the acetic acid by adding 0.010 g for the 50 ml used.

5. CALCULATION

5.1 Calculate the percentage of ethyl centralite in the sample as follows:

$$\text{Percentage of Ethyl Centralite} = \frac{(A-B)62.91}{W}$$

where

A = total wt of brominated products
B = corrected weight of tetrabromdiphenylamine
W = weight of sample

5.2 If the powder was made with methyl centralite use the following calculation:

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$$\text{Percentage of Methyl Centralite} = \frac{(A-B)60.30}{W}$$

where

A = total wt of brominated products
B = corrected weight of tetrabromdiphenylamine
W = weight of sample

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Method No. 203.1

DIETHYLPHTHALATE (CHROMATOGRAPHIC-GRAVIMETRIC METHOD)

1. APPARATUS

1.1 Prepare a chromatographic column as shown in figure 10 or by fitting a piece of pyrex (or equal) thick-walled glass tubing approximately 28 mm inside diameter by 5 inches long into a 30 ml size fritted glass funnel containing a small cushion of glass wool. Make the connection between the tube and the funnel by means of a piece of Gooch type tubing. Fit the funnel into a 500-ml side arm suction flask which is connected to a water aspirator. With the aspirator on full force, add a 2:1 silicic acid-Celite (Merck or equal) adsorbent to the column by means of a spatula. When the column is full, firmly push down the adsorbent with the aid of a cork which barely fits into the tube. Fit the top of the tube with a stopper bearing a 200-ml separatory funnel and a piece of capillary tubing extending from a point above half the height of the separatory funnel to reach 1/4 inch into the tube. The cork shall make an airtight seal with the column.

2. PROCEDURE

2.1 Adsorption.--Dissolve one of the dried residues from the ether extract determination (obtained as directed in Method No. 104.1) in sufficient 1:1 benzene-ligroin solution to make a volume of 40 ml (dissolve the sample in benzene and then add ligroin). Pass, in order, the following solutions into the column while the latter is under full suction of the water aspirator: (The solutions shall not touch the cork nor shall the solution be allowed to be sucked below the top of the adsorbent between additions.)

85 ml anhydrous ether
85 ml ligroin
40 ml benzene-ligroin solution of the sample prepared above
5 rinsings of the sample flask with 5 ml portions of benzene-ligroin
175 ml benzene-ligroin
85 ml ligroin

Maintain the vacuum for 5 minutes after the final ligroin has passed into the top of the column.

2.2 Solution, precipitation and calculation.--Remove the adsorbent with a wooden rod which just fits into the tube. Discard the thin, colored layer at the very top of the column (about 1 mm thick) and transfer about 3 1/4 inches of the column, immediately below the colored layer, to a 100-ml fritted glass filtering funnel fitted to a bell jar filtration apparatus. Add 40 ml of dry ether to the adsorbent in the funnel. Mix the adsorbent with the ether by stirring with a glass rod. Remove the ether by applying suction and catch the filtrate in a 250 ml beaker. Repeat the extraction with three more 40 ml portions of dry ether. Evaporate the ether at about 40°C to about 50 ml with the aid of a gentle current of dry air, then transfer the ether to a ground glass stoppered 125 ml. Erlenmeyer flask. Evaporate the ether and dry the residue for 15 minutes in a vacuum desiccator.

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Make sure that no moisture remains in the flask. Pipette into the flask, 10 ml of a previously filtered 0.5N solution of potassium hydroxide in absolute alcohol. Loosely stopper the flask and heat for 1 hour on an open steam bath. After the contents of the flask have been heated to the temperature of the bath, release the pressure in the flask and tightly close the glass stopper. After 1 hour, remove the flask from the bath, loosen the stopper and cool in a water bath to room temperature. Transfer the precipitate of dipotassium phthalate onto a tared asbestos mat crucible and wash with 10-20 ml of absolute alcohol. Particles in the flask may be transferred with the use of a rubber policeman. Dry the crucible at 150°C for 2 hours and weigh. Calculate the percent diethylphthalate as follows:

$$\text{Percent diethylphthalate} = \frac{91.7\ (A - 0.0100\ g)}{W}$$

where

A = gram of precipitate
W = gram of sample corrected for volatiles

2.3 Correction factor.--The correction factor of 0.0100 g, applied in the calculation of the percent of diethylphthalate in the above calculation is the amount of potassium nitrate derived from the organic nitrate impurities present in the nitroglycerin by decomposition with alcoholic potassium hydroxide. This factor should be checked frequently as follows: Dissolve the precipitate from the filtering crucible by sucking five 10 ml portions of water through the crucible into a 150 ml side arm suction flask. Quantitatively transfer the filtrate to a 100 ml volumetric flask and dilute to the mark. Transfer a 5 ml aliquot to a 50 ml beaker by means of a pipette. Evaporate the 5 ml to dryness on a steam bath. Cool the beaker, and add 2 ml of phenoldisulfonic acid reagent. (See below) Mix thoroughly with a stirring rod to insure intimate contact, and after 5 minutes, dilute with 25 ml of water. Neutralize the solution with 1:1 ammonium hydroxide and continue to add ammonium hydroxide until the maximum color has been developed. Transfer the solution to a 50 ml volumetric flask and dilute to the mark. Determine the amount of potassium nitrate present colorimetrically by comparison with a standard potassium nitrate solution containing 0.100 g of potassium nitrate in 1 liter of water. A 5 ml aliquot of the standard solution is treated in a similar manner as the unknown solution.

2.4 Preparation of phenoldisulfonic acid reagent.--Dissolve 6 g of phenol in 37 ml of concentrated H_2SO_4. Heat on a steam bath until solution is complete. Cool and add 3 ml of water.

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Method No. 204.1

DIBUTYLPHTHALATE (DIFFERENCE METHOD)

1. PROCEDURE

1.1 Subtract from the percentage of ether extractive matter the sum of the percentages of diphenylamine, dinitrotoluene, and nitroglycerin and report the difference as percentage of dibutylphthalate on a moisture and volatile solvents-free basis.

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Method No. 204.2

DIBUTYLPHTHALATE (GRAVIMETRIC)

1. PROCEDURE

1.1 Weigh accurately a 5-gm portion of the powder, using sliced grains of the larger powders. Place this portion in a 500-ml flask fitted to a reflux condenser by means of a ground-glass joint, and add 40 ml of approximately 35-percent nitric acid solution. Reflux the mixture vigorously for about 45 minutes until the decomposition is complete and clear solution is obtained. Continue refluxing for 30 minutes longer to ensure complete reaction. Do not interrupt this operation, or a low value will be obtained. Allow the mixture to cool and add approximately 50 ml of water. Remove the flask, insert a glass stopper, and shake until nitrodiphenylamine and dinitrotoluene begin to settle out. Allow to stand overnight if possible. Filter the mixture through a Whatman No. 42 or equivalent filter paper, catch the filtrate in a 400-ml beaker, and wash the solids on the filter three or four times with water. If the volume of the filtrate and washings exceeds 100 ml, evaporate the solution to this volume on a hot plate. Place the beaker on a steam bath and remove any oxalic acid in solution by adding potassium permanganate a few crystals at a time, and allowing the reaction to complete itself before the next addition. Continue this until a large precipitate of manganese dioxide persists and apparently does not dissolve when heating on the steam bath is continued. (Although about 3.3 gm of potassium permanganate will be required this must be added only in quantities of a few crystals at a time until a large precipitate persists and not just to the appearance of a pink color). (See note for procedure when potassium sulfate is present.) Filter the solution while hot through a Whatman No. 42 or equivalent filter paper and catch the filtrate in a 250-ml beaker. Wash the precipitate four times with hot water and catch the washings in the same beaker. Evaporate the filtrate slowly to a volume of about 5 ml on a hot plate and then evaporate to dryness on a steam bath so as to remove completely nitric acid, in which lead phthalate is soluble. Dissolve the phthalic acid and manganese salts in the beaker by adding 50 ml of warm water. Add 20 ml of a 10-percent lead acetate solution containing 10 ml of acetic acid per 100 ml of solution. Evaporate on a hot plate, using a low heat, until the volume of the solution has been reduced to about 50 ml and allow to stand overnight. Filter the solution through a Whatman No. 42 or equivalent filter paper and wash the white crystalline precipitate of lead phthalate three or four times with water from the beaker onto the filter. Keep the volume of the washings at a minimum of about 20 ml. In some cases, lead phthalate crystals will adhere to the beaker. Do not attempt to remove these, but allow them to remain in the beaker. Add 15 ml of approximately 35-percent nitric acid solution to the beaker to dissolve any crystals present, pour the solution from the beaker onto the lead phthalate, and catch the resulting lead nitrate solution in a 400-ml beaker. When the solution of the lead phthalate is practically complete, puncture the filter paper with a stirring rod and wash the contents into the 400-ml beaker. Add 10 ml of approximately 95-percent sulfuric acid solution to precipitate any lead present as lead sulfate. Evaporate the solution to fumes of sulfur trioxide, cool, and add 100 ml of approximately 47.5-percent (by volume) alcohol solution. Stir

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the mixture, allow to stand for 2 hours, filter through a tared filtering crucible, and wash with approximately 47.5-percent (by volume) alcohol solution. Dry the crucible and contents for 3 hours in an oven at 110°C, cool in a desiccator, and weigh. Calculate the weight of lead sulfate to percentage of dibutylphthalate in the powder on a moisture and volatile solvents-free basis as follows:

$$\text{Percentage of dibutylphthalate} = \frac{91.66A}{W}$$

where

A = weight of residue
W = weight of sample on a moisture and volatile solvents-free basis

1.2 NOTE on dibutylphthalate determination: When potassium sulfate is present filter off the manganese dioxide while the solution is hot into a 400-ml beaker, and wash with hot water. Place the beaker on a hot plate and slowly evaporate nearly to dryness then bring to dryness on the steam bath. When crystals start to form add 5 ml of distilled water and leave on the steam for 3 hours. Then remove, add 100 ml of water and heat to boiling. Add 50 ml of Ba $(NO_3)_2$ solution, (3.5 gm Ba $(NO_3)_2$ to a liter of H_2O) continue to boil for 30 minutes, and finally filter off $BaSO_4$ formed while solution is still hot. Catch the filtrate quantitatively in a clean filter flask, washing the precipitate with four or five portions of hot water. Transfer contents of filter flask carefully back into original beaker, and evaporate to a volume of 30 ml. Add 20 ml of lead acetate solution and complete the determination in the regular manner.

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Method No. 205.1

DINITROTOLUENE (Ester Nitrates Absent)

1. REAGENTS

1.1 Preparation of solution.--Prepare the volumetric solutions that are necessary for this determination with freshly distilled water which is free from atmospheric oxygen. Connect the siphon tubes and burettes in such a way that only carbon dioxide gas, supplied from a Kipp generator, will be drawn into the stock bottles as the solutions are used. Replace the rubber tubing used for connections whenever cracks appear. When no titrations are being made, turn off the stopcock in the tube connecting the Kipp generator with the stock bottle of titanous chloride solution and the burette (Figure 11).

1.2 Titanous chloride solution.--Prepare a 0.2 normal solution of titanous chloride by mixing 150 ml of 20 percent titanium trichloride solution with 100 ml of 38 percent hydrochloric acid solution and diluting the resulting solution to 1 liter with distilled water. Keep the solution protected from the air as much as possible by means of carbon dioxide. Mix the solution thoroughly with a current of carbon dioxide and store in a bottle covered with black paint or black paper to exclude light.

1.2.1 Standardization[a].--Transfer an accurately weighed portion of 3.0 to 3.5 g of pure hexahydrated ferrous ammonium sulfate of known iron content to a 400 ml beaker and dissolve in 50 to 75 ml of water. Add 10 ml of 38 percent hydrochloric acid solution and 0.5 g of potassium chlorate, then evaporate to dryness on a steam bath. Add 50 ml of water, 5 ml of 38 percent hydrochloric acid solution and evaporate again to dryness. Pass a current of carbon dioxide through a special type titration flask (Figure 12) for five minutes to displace the air. Dissolve the resulting ferric salt in a minimum quantity of water and transfer to the flask. Add 25 ml of 15 percent hydrochloric acid solution and titrate with 0.2 normal titanous chloride solution, keeping a current of carbon dioxide passing through the flask. Near the end point, which can be judged by the gradual disappearance of the reddish-colored ferric salt, add 5 ml of 20 percent ammonium thiocyanate solution. Continue the titration until the blood-red color of ferric thiocyanate just disappears. Apply temperature and burette corrections to the observed reading.

$$\text{Normality of titanous chloride solution} = \frac{W}{0.3921\ V}$$

where

W = weight of ferrous ammonium sulfate
V = corrected volume of titanous chloride solution

Check the normality of the titanous chloride solution at least once a week by titration with ferric ammonium sulfate solution.

1.3 Ferric ammonium sulfate solution (0.15N).--Prepare a 0.15 normal ferric alum solution by mixing 75 grams of hydrated ferric ammonium sulfate

[a] Titanous chloride solution may be standardized by using U.S. Bureau of Standards Potassium Dichromate, No. 136, and 0.5 percent aqueous solution of sodium diphenylbenzidine sulfonate as indicator. In case of dispute, use the method of standardization described above.

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($Fe_2(SO_4)_3 \cdot (NH_4)_2SO_4 \cdot 24H_2O$) with 25 ml of 95 percent sulfuric acid solution for each liter of solution, and mix thoroughly by means of a current of carbon dioxide. Displace the air in a titration flask with carbon dioxide. Accurately measure 40 to 45 ml of ferric ammonium sulfate solution into the flask, add 25 ml of 15 percent hydrochloric acid solution and 50 ml of water. Titrate with 0.2 normal titanous chloride solution until near the end point, then add 5 ml of 20 percent ammonium thiocyanate solution. Continue the titration until the red color just disappears. Apply temperature and burette corrections to the observed readings and calculate the normality of the ferric ammonium sulfate solution. It can be assumed that the strength of the solution will remain constant for an indefinite period.

2. DETERMINATION

2.1 Dissolve the dried residue from the determination of ether extractive matter obtained as directed in Method No. 104.1 with glacial acetic acid, transfer the solution to a 250 ml volumetric flask, make up to volume with glacial acetic acid and mix thoroughly. Avoid temperature changes on account of the high thermal coefficient of expansion of acetic acid. Displace the air in a titration flask by passing in a current of carbon dioxide for five minutes. Rinse a calibrated 25 ml pipette with the acetic acid solution of the sample, then transfer a 25 ml portion of the solution to the flask. Add an accurately measured excess of 0.2 normal titanous chloride solution (4 ml for each 1 percent of dinitrotoluene present) and 25 ml of 15 percent hydrochloric acid solution, connect the flask to a reflux condenser, and boil gently for five minutes on a hot plate. Add a few glass beads to prevent bumping. Increase the current of carbon dioxide, then cautiously immerse the flask in a large beaker of cold water, keeping the index finger over the top of the condenser until the hot vapors are condensed. After cooling the flask and contents to room temperature, disconnect the condenser, and insert a one-hole rubber stopper which contains a short piece of glass tubing. Insert the tip of the burette into the glass tubing, and titrate the cooled solution with 0.15 normal ferric ammonium sulfate solution to the first permanent red color. Run a blank, following the above procedure, to correct for reducible impurities in the reagents. Sweep the air from a titration flask, add 25 ml of glacial acetic acid, 25 ml of 15 percent hydrochloric acid solution, and about 20 ml of 0.2 normal titanous chloride solution. Connect the flask to the condenser and boil the solution for five minutes on the hot plate. Cool the flask and contents, and titrate with 0.15 normal ferric ammonium sulfate solution, adding 5 ml of 20 percent ammonium thiocyanate solution near the end point. Calculate the percentage of Dinitrotoluene.

$$\text{Percentage of Dinitrotoluene} = \frac{1.517\ (V - AF)\ (N)}{W}$$

Where

V = volume of titanous chloride solution added
N = normality of titanous chloride solution

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A = volume of ferric ammonium sulfate solution
W = weight of sample on total volatile free basis
F = factor for converting volume of ferric ammonium sulfate solution into volume of titanous chloride solution obtained for the blank

In the above calculation the molecular weight of dinitrotoluene is divided by twelve to obtain the equivalent weight.

Method No. 205.2

DINITROTOLUENE (ESTER NITRATES PRESENT)

1. REAGENTS

1.1 Use reagents described under Method 205.1.

2. DETERMINATION

2.1 Proceed as directed under Nitroglycerin, Method 208.1, paragraph 2.1, then continue as directed in paragraph 2.1, above, beginning with line 9.

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Method No. 206.1

TRINITROTOLUENE (TITRATION METHOD) (Ester Nitrates Absent)

1. REAGENTS

1.1 Preparation of solution.--Prepare the volumetric solutions that are necessary for this determination with freshly distilled water which is free from atmospheric oxygen. Connect the siphon tubes and burettes in such a way that only carbon dioxide gas, supplied from a Kipp generator, will be drawn into the stock bottles as the solutions are used. Replace the rubber tubing used for connections whenever cracks appear. When no titrations are being made, turn off the stopcock in the tube connecting the Kipp generator with the stock bottle of titanous chloride solution and the burette (Figure 11).

1.2 Titanous chloride solution.--Prepare a 0.2 normal solution of titanous chloride by mixing 150 ml of 20 percent titanium trichloride solution with 100 ml of 38 percent hydrochloric acid solution and diluting the resulting solution to 1 liter with distilled water. Keep the solution protected from the air as much as possible by means of carbon dioxide. Mix the solution thoroughly with a current of carbon dioxide and store in a bottle covered with black paint or black paper to exclude light.

1.2.1 Standardization[a].--Transfer an accurately weighed portion of 3.0 to 3.5 g of pure hexahydrated ferrous ammonium sulfate of known iron content to a 400 ml beaker and dissolve in 50 to 75 ml of water. Add 10 ml of 38 percent hydrochloric acid solution and 0.5 g of potassium chlorate, then evaporate to dryness on a steam bath. Add 50 ml of water, 5 ml of 38 percent hydrochloric acid solution and evaporate again to dryness. Pass a current of carbon dioxide through a special type titration flask (Figure 12) for five minutes to displace the air. Dissolve the resulting ferric salt in a minimum quantity of water and transfer to the flask. Add 25 ml of 15 percent hydrochloric acid solution and titrate with 0.2 normal titanous chloride solution, keeping a current of carbon dioxide passing through the flask. Near the end point, which can be judged by the gradual disappearance of the reddish-colored ferric salt, add 5 ml of 20 percent ammonium thiocyanate solution. Continue the titration until the blood-red color of ferric thiocyanate just disappears. Apply temperature and burette corrections to the observed reading.

$$\text{Normality of titanous chloride solution} = \frac{W}{0.3921\ V}$$

where

W = weight of ferrous ammonium sulfate
V = corrected volume of titanous chloride solution

Check the normality of the titanous chloride solution at least once a week by titration with ferric ammonium sulfate solution.

1.3 Ferric ammonium sulfate solution (0.15N).--Prepare a 0.15 normal ferric alum solution by mixing 75 grams of hydrated ferric ammonium sulfate

a Titanous chloride solution may be standardized by using U.S. Bureau of Standards Potassium Dichromate, No. 136, and 0.5 percent aqueous solution of sodium diphenylbenzidine sulfonate as indicator. In case of dispute, use the method of standardization described above.

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$(Fe_2(SO_4)_3 \cdot (NH_4)_2SO_4 \cdot 24H_2O)$ with 25 ml of 95 percent sulfuric acid solution for each liter of solution, and mix thoroughly by means of a current of carbon dioxide. Displace the air in a titration flask with carbon dioxide. Accurately measure 40 to 45 ml of ferric ammonium sulfate solution into the flask, add 25 ml of 15 percent hydrochloric acid solution and 50 ml of water. Titrate with 0.2 normal titanous chloride solution until near the end point, then add 5 ml of 20 percent ammonium thiocyanate solution. Continue the titration until the red color just disappears. Apply temperature and burette corrections to the observed readings and calculate the normality of the ferric ammonium sulfate solution. It can be assumed that the strength of the solution will remain constant for an indefinite period.

2. DETERMINATION

2.1 Dissolve the dried residue from the determination of ether extractive matter (obtained as directed in Method No. 104.1) with glacial acetic acid, transfer the solution to a 250 ml volumetric flask, make up to volume with glacial acetic acid and mix thoroughly. Avoid temperature changes on account of the high thermal coefficient of expansion of acetic acid. Displace the air in a titration flask by passing in a current of carbon dioxide for five minutes. Rinse a calibrated 25 ml pipette with the acetic acid solution of the sample, then transfer a 25 ml portion of the solution to the flask. Add an accurately measured excess of 0.2 normal titanous chloride solution (4 ml for each 1 percent of dinitrotoluene present) and 25 ml of 15 percent hydrochloric acid solution, connect the flask to a reflux condenser and boil gently for five minutes on a hot plate. Add a few glass beads to prevent bumping. Increase the current of carbon dioxide, then cautiously immerse the flask in a large beaker of cold water, keeping the index finger over the top of the condenser until the hot vapors are condensed. After cooling the flask and contents to room temperature, disconnect the condenser, and insert a one-hole rubber stopper which contains a short piece of glass tubing. Insert the tip of the burette into the glass tubing, and titrate the cooled solution with 0.15 normal ferric ammonium sulfate solution to the first permanent red color. Run a blank, following the above procedure, to correct for reducible impurities in the reagents. Sweep the air from a titration flask, add 25 ml of glacial acetic acid, 25 ml of 15 percent hydrochloric acid solution, and about 20 ml of 0.2 normal titanous chloride solution. Connect the flask to the condenser and boil the solution for five minutes on the hot plate. Cool the flask and contents, and titrate with 0.15 normal ferric ammonium sulfate solution, adding 5 ml of 20 percent ammonium thiocyanate solution near the end point.

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$$\text{Percentage of Trinitrotoluene} = \frac{1.261\ (V - AF)\ (N)}{W}$$

where

V = volume of titanous chloride solution added
N = normality of titanous chloride solution
A = volume of ferric ammonium sulfate solution
W = weight of sample on total volatile free basis
F = factor for converting volume of ferric ammonium sulfate solution into the volume of titanous chloride solution obtained for the blank

In the above calculation the molecular weight of Trinitrotoluene is divided by eighteen to obtain the equivalent weight.

Method No. 206.2

TRINITROTOLUENE (DIFFERENCE METHOD)

1. PROCEDURE

1.1 Calculate the percentage of Trinitrotoluene present in the powder by subtracting from the weight of ether extractive matter the sum of weights of triacetin and diphenylamine found present and dividing by the weight of the sample on a moisture and volatile free basis.

Method No. 206.3

TRINITROTOLUENE (ESTER NITRATES PRESENT)

1. PROCEDURE

1.1 Proceed as directed in Method No. 205.2 using the calculation given in Method No. 206.1, paragraph 2.1.

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TRIACETIN (EXTRACTION METHOD)

1. PROCEDURE

1.1 Wash the dried and weighed ether extractive matter in the extraction flask with eight 25-cc. portions of distilled water saturated with TNT at room temperature. Shake the contents of the flask with each 25-cc portion to obtain the maximum washing effect. Decant each portion of water through a 12 cm filter (Whatman No. 42 preferably) so as to transfer as little as possible of the undissolved solids to the filter. After the last water-wash has passed through the filter, place the extraction flask under the funnel, perforate the tip of the filter paper, and wash any solids held on the filter into the extraction flask by means of several 1-cc portions of acetone. Evaporate the acetone from the flask by means of a current of dry air and dry the flask and contents in a sulphuric acid vacuum desiccator for 16 hours. Determine the loss in weight of the contents of the flask and calculate this as percentage of triacetin in the powder on a moisture and volatile free basis.

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Method No. 208.1

NITROGLYCERIN (TITRATION)

1. PREPARATION OF SOLUTIONS

1.1 Prepare the volumetric solutions that are necessary for this determination with freshly distilled water which is free from atmospheric oxygen. Connect the siphon tubes and burets in such a way that only carbon dioxide gas, supplied from a Kipp generator, will be drawn into the stock bottles as the solutions are used. Replace the rubber tubing used for connections whenever cracks appear. When no titrations are being made, turn off the stopcock in the tube connecting the Kipp generator with the stock bottle of titanous chloride solution (Fig. 11).

1.2 Titanous chloride solution.--Prepare a 0.2-normal solution of titanous chloride by mixing 150 ml of 20-percent titanium trichloride solution with 100 ml of 38-percent hydrochloric acid solution and diluting the resulting solution to 1 liter with distilled water. Keep the solution protected from the air as much as possible by means of carbon dioxide. Mix the solution thoroughly with a current of carbon dioxide and store in a bottle covered with black paper to exclude light.

1.2.1 Standardization[a].--Transfer an accurately weighed portion of 3.0 to 3.5 grams of pure hexahydrated ferrous ammonium sulphate of known iron content to a 400 ml beaker and dissolve in 50 to 75 ml of water. Add 10 ml of 38-percent hydrochloric acid solution and 0.5 gram of potassium chlorate, then evaporate to dryness on a steam bath. Add 5 ml of water, 5 ml of 38-percent hydrochloric acid solution and evaporate again to dryness. Dissolve and transfer the residue with about 50 ml of water to a special titration flask (See Fig. 12) through which carbon dioxide gas has been passing for 5 minutes to displace the air. Add 25 ml of 15-percent hydrochloric acid solution and titrate with 0.2-normal titanous chloride solution, keeping a current of carbon dioxide passing through the flask. Near the end point, which can be judged by the gradual disappearance of the red-colored ferric salt, add 5 ml of 20-percent ammonium thiocyanate solution. Continue the titration until the blood-red color of ferric thiocyanate just disappears. Apply temperature and buret corrections to the observed reading.

$$\text{Normality of titanous chloride solution} = \frac{A}{0.39214B}$$

where

A = weight of ferrous ammonium sulphate
B = corrected volume of titanous chloride solution

Check the normality of the titanous chloride solution at least once a week by means of a cross titration with ferric ammonium sulfate solution.

1.3 Ferric ammonium sulfate solution (0.15N).--Prepare a 0.15-normal ferric ammonium sulfate solution by dissolving 75 grams of hydrated ferric ammonium sulfate ($Fe_2(SO_4)_3 \cdot (NH_4)_2SO_4 \cdot 24H_2O$) with 25 ml of 95-percent sulphuric acid solution in distilled water and diluting to 1 liter. Mix

a Titanous chloride solution may be standardized by using U.S. Bureau of Standards Potassium Dichromate, No. 136, and 0.5 percent aqueous solution of sodium diphenylbenzidine sulfonate as indicator. In case of dispute use the method of standardization described above.

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thoroughly by means of a current of carbon dioxide gas. Displace the air in a titration flask with carbon dioxide.

1.3.1 Standardization.--Accurately measure 40 to 45 ml of ferric ammonium sulfate solution into the flask and add 25 ml of 15-percent hydrochloric acid solution. Titrate with 0.2-normal titanous chloride solution until near the end point, then add 5 ml of 20-percent ammonium thiocyanate solution. Continue the titration until the red color just disappears. Apply temperature and buret corrections to the observed readings and calculate the normality of the ferric ammonium sulfate solution. It can be assumed that the strength of the solution will remain constant for an indefinite period.

$$\text{Normality of ferric ammonium sulfate solution} = \frac{AN}{B}$$

where

A = ml of titanous chloride solution used
N = normality of titanous chloride solution
B = ml of ferric ammonium sulfate solution used

1.4 Ferrous chloride solution (0.7N).—Prepare a 0.7-normal ferrous chloride solution by dissolving 140 gm of ferrous chloride tetrahydrate in 50 ml of 38 percent hydrochloric acid and approximately 800 ml of water and diluting the resulting solution to 1 liter with water. Mix the solution thoroughly by means of a current of carbon dioxide. This solution invariably contains a small amount of ferric iron, which must be determined by running a blank under the same conditions as in the analysis for Nitroglycerin. Displace the air from a special type titration flask (See Fig.12) with a current of carbon dioxide. Add an accurately measured portion of 15 ml of 0.7-normal ferrous chloride solution, 25 ml of 15-percen hydrochloric acid solution, 25 ml of glacial acetic acid, and boil for 5 minutes under a reflux condenser on a hot plate. Cool to room temperature by means of a large beaker of cold water, increasing the current of carbon dioxide so that no air will be sucked down into the flask. It is convenient to hold the index finger over the top of the condenser until equilibrium is established. Disconnect the condenser and add 5 ml of 20-percent ammonium thiocyanate solution, then titrate with 0.2 normal titanous chloride solution until the red color just disappears. This volume of titanous chloride solution, usually 0.5 to 1.0 ml must be subtracted from subsequent titrations involved in the determination of Nitroglycerin. Run a blank at frequent intervals to determine whether the ferrous chloride solution has changed. Whenever any new bottles of reagents are used, run a new blank to detect any interfering impurities in these reagents.

2. PROCEDURE

2.1 Dissolve the dried residue from the determination of ether extractive matter with glacial acetic acid, transfer the solution to a 250 ml calibrated volumetric flask, make up to volume the glacial acetic acid and mix thoroughly. Avoid temperature changes on account of the high thermal

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coefficient of expansion of acetic acid. Displace the air in a titration flask by passing in a current of carbon dioxide for 5 minutes. Rinse a calibrated 25 ml pipette with the acetic acid solution of the sample, then transfer a 25 ml portion of the solution to the flask. Add an accurately measured portion of 15 ml of 0.7-normal ferrous chloride solution and 25 ml of 15-percent hydrochloric acid solution, connect the flask to a reflux condenser, and boil the solution gently for five minutes on a hot plate. Add a few glass beads to prevent possible bumping. The color of the solution will change from yellow to dark green to reddish-brown as the reaction proceeds. Increase the current of carbon dioxide, then cautiously immerse the flask in a larger beaker of cold water, keeping the index finger over the top of the condenser until the hot vapors are condensed. After cooling the flask and contents to room temperature, disconnect the condenser, and titrate with 0.2 normal titanous chloride solution until near the end point, as shown by the disappearance of the reddish-brown color of ferric iron. Add 5 ml of 20 percent ammonium thiocyanate solution and continue the titration until the deep red color of ferric thiocyanate disappears. Apply temperature and burette corrections to the observed reading. Calculate the nitroglycerin content on a volatiles-free basis as follows:

$$\text{Percentage of nitroglycerin} = \frac{2.523(V-v)\,N}{W}$$

where

V = volume of titanous chloride solution used.
v = volume of titanous chloride solution used in blank
N = normality of titanous chloride solution
W = weight of sample corrected for volatiles, represented by the aliquot taken

The titration on duplicate aliquot portions shall check within 0.05 ml. In the above calculation, the molecular weight of nitroglycerin is divided by 9 to obtain the equivalent weight.

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Method No. 209.1

NITROCELLULOSE (PRECIPITATION METHOD)

1. PROCEDURE

1.1 Prepare at least 20 g of powder by cutting thin slices from or crushing 20 or more grains. Weigh accurately three portions of the crushed powder, one of three grams and two of one gram each, and wrap them separately in filter paper. Place the three portions in an extraction thimble or crucible and extract in a Soxhlet or other suitable extraction apparatus, the flask of which has been tared previously. Continue the extraction until complete as indicated by absence of residue when a portion of the ether is evaporated to dryness. The time required is approximately 120 hours. Retain the ether extract for the determination of extractive matter. Place each of the extracted 1 gram portions of the sample in a tared aluminum can about 3-1/2 inches in diameter and 2 inches deep, fitted with a cover and stirring rod. Add 50 cc of alcohol complying with Grade 2 of Specification No. JAN-A-463 and 100 cc of ether complying with Specification No. JAN-E-199; stir, cover, and leave can under a bell jar overnight. Replace the ether that has evaporated and stir the contents until the nitrocellulose is in solution. Place the can on a closed steam bath and evaporate a portion of the ether. (The amount of ether evaporated before the precipitation of the pyrocellulose determines the character of the precipitate; if too much ether has evaporated a gummy precipitate will be obtained, while if too little has evaporated, the precipitate will be sandy and cause bumping. The amount of ether to evaporate in order to obtain a fine, flaky precipitate can be learned only by practice.) When the proper amount of ether has been evaporated, remove the can from the steam bath and add 5 cc of distilled water with constant stirring. Add 45 cc more water in small portions stirring the contents of the can throughout the addition. If the precipitate seems to be too heavy add 5 cc portions of ether and stir until the precipitate becomes light and flaky. Replace the can on the steam bath and stir continuously until the bulk of the ether has been evaporated and there is no danger of bumping. Take the residue just to dryness and then dry for 1-1/2 hours at 100°C. Cover, cool in a desiccator and weigh rapidly. Calculate the weight of residue to percentage of nitrocellulose in the powder as received and also on a moisture and volatile-free basis.

Method No. 209.2

NITROCELLULOSE (DIFFERENCE METHOD)

1. PROCEDURE

1.1 The percent nitrocellulose shall be calculated by subtracting from 100 percent, the sum of the percentage of dinitrotoluene, diphenylamine, potassium sulfate, tin, ash, graphite, and any other compounding ingredients which may be present. The graphite percentage which is usually very small shall be furnished by the contractor on the basis of the manufacturing data.

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Method No. 209.3

NITROGEN IN NITROCELLULOSE

1. NITROCELLULOSE

1.1 To obtain a stock of nitrocellulose for a determination of nitrogen, acetone, or ether-alcohol solubility, viscosity and other items, dissolve 20 grams of powder, previously cut into small pieces or ground and pressed through a 15 mesh sieve, in acetone to make circa one percent solution. Warm one liter of distilled water for each 5 grams of powder in solution in acetone and precipitate the nitrocellulose by slowly adding the powder solution to the four separate liters of water. Allow to settle and siphon off the water layer. Wash the residues into smaller beakers, evaporate dry on a water bath. Remove from beakers with clean tweezers to paper trays. Pull the fibers apart and pull any lumps apart. Diphenylamine is present and must be removed by extraction for 2 - 3 hours in a Soxhlet or Knorr with ether (alcohol free). Dry an ether extracted nitrocellulose for one hour at 100°C. Caution: Do not put in hot oven until ether has evaporated as it may flash.

2. NITROGEN CONTENT

2.1 (1). Nitrometer method.--

2.1.1 (1) a. Standardization of the nitrometer.--Standardization shall be made by either of the following methods:

2.1.2 Absolute method.--Assemble the nitrometer (duPont 5 part form of Lunge's nitrometer, or equivalent) after thoroughly cleaning and drying all parts. The various parts, with the exception of the reservoir bulbs, shall be completely filled with mercury. Place 25 ml of 94.5 $\pm$ 0.5 percent sulfuric acid in the generating bulb, draw air into the bulb to nearly the level of the lower shoulder by lowering the mercury reservoir and opening both stopcocks. Close both stopcocks and shake vigorously for 3 minutes. Connect the generating bulb to the compensating tube. Draw the dry air from the generating bulb into the compensating tube by adjustment of the 2 mercury reservoirs, keeping open the stopcock of the measuring tube. Continue the introduction of the air until the mercury in the reading tube is nearly at the 12.50 mark. Seal the compensating tube by means of a small blow pipe flame. Desiccate similarly in the generating bulb another portion of air and transfer this to the reading tube until the mercury in the reading tube, compensating tube, and their mercury reservoirs is at the same level, the mercury in the reading tube being approximately at the 12.50 mark. It is not necessary that these gas volumes be exact or equal. Fill a small U-tube manometer with sulfuric acid and attach to the reading tube. Allow the apparatus to come to constant temperature and then adjust the mercury levels in the reading tube, compensating tube, and their mercury reservoirs so that the air in the reading tube is under atmospheric pressure, as shown by the acid levels in the arms of the manometer tube. Take the reading on the graduated scale of the calibrated reading tube, and at the same time note the temperature and barometric pressure. From these readings calculate the volume of air at 20°C and 760 mm pressure in the reading tube by means of the following formula:

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$$V = V' \times \frac{P' \times 293 \times (1-0.00018t')}{760 \times (273 + T') \times (1-0.00018 \times 20)}$$

where

V = volume at 20°C and 760 mm barometer at 20°C
V' = volume at the observed temperature (T') and pressure (P')
t' = temperature of the barometer

The factor 0.00018 represents the correction to be applied for the coefficient of expansion of mercury in the barometer. When V has been calculated, adjust the heights of the reading tube, compensating tube, and their reservoirs so that, with the mercury in the reading and compensating tubes on a level, the mercury in the reading tube marks the calculated volume of the air. Paste a strip of paper on the tube so as to mark the height of the mercury in the compensating tube. Take this as the standard volume of air with which every volume of gas to be measured is compared.

NOTE.--The use of dry air tends to form a scum around the reading level of the compensating tube, thus requiring frequent restandardization of the nitrometer. This scum is caused by reaction of oxygen in the air with mercury or by impurities. Dry nitrogen eliminates this condition and should be used, if available. A half drop of sulfuric acid will also prevent this scum.

2.1.3 Potassium nitrate method.--Recrystallize 3 times from 95 percent alcohol a sample of chemically pure grade of potassium nitrate, and grind this to pass through a No. 100 U. S. Standard sieve in accordance with Federal Specification RR-S-366. After drying at 135° to 150°C for 2 or 3 hours, weigh a portion of exactly 1 gm into a small weighing bottle and transfer this portion to the cup of the generating bulb. Wash the potassium nitrate into the generating bulb by means of 8 to 10 portions of 94.9 ± 0.1 percent sulfuric acid free from nitrogen, using a total of 25 ml of acid. With the lower stopcock of the generating bulb open and the mercury reservoir low enough to maintain a slightly reduced pressure in the generating bulb, shake the sample until most of the gas has been generated. Adjust the mercury reservoir until the mercury drops nearly to the lower shoulder of the generating bulb. Close the lower stopcock and shake the bulb vigorously for 3 minutes. Replace the bulb on the rack, open the lower stopcock, and allow the tube to stand for 5 minutes, the levels in the generating bulb and reservoir being approximately the same having previously introduced desiccated air into the compensating tube as in the absolute method described in paragraph 2.1.2. Close the lower stopcock, and shake the generating bulb for an additional minute. Transfer the gas to the reading tube, and adjust the levels of the mercury in the reading and compensating tubes to approximately the same height. Allow the gas to stand for approximately 20 minutes to permit equalization of the temperature. With sulfuric acid U-tube leveling device, make careful adjustment so that the mercury in both tubes is at the same level and that in the reading tube is at the 13.85 mark (the percentage of nitrogen in potassium nitrate). Make a mark on the compen-

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sating tube to show the volume occupied by the air equivalent to the volume of nitric oxide gas liberated from exactly 1 gm of potassium nitrate. This mark is usually made on a strip of gummed paper attached to the compensating tube.

NOTE 1.--In order to avoid any doubt regarding the purity of the potassium nitrate, it is desirable to check the standardization obtained by the use of potassium nitrate by means of the absolute method described in paragraph 2.1.2. If the agreement is within 1 or 2 hundredths of a percent the potassium nitrate can be considered as of satisfactory quality and used in subsequent periodic standardizations. It has been found by actual experimentation that the solubility of nitric oxide in sulfuric acid of the strength prescribed is so small that no correction need be applied to bring the value in agreement with the mark set by the absolute method described in paragraph 2.1.2.

NOTE 2.--If the nitrometer is standardized by both the absolute and potassium nitrate methods described in paragraphs 2.1.2 and 2.1.3, the results may not be in agreement by 0.01 to 0.04 percent nitrogen due to expansion of the brass barometer scale. Such difference may be removed by applying the brass scale correction corresponding to the temperature as described in Lange's Handbook of Chemistry, Page 1605, 5th edition, 1944.

3. PROCEDURE

3.1 Air-dry a portion of the sample at a temperature not above 46°C and transfer appoximately 1 gm of the material to a ground-glass stoppered weighing bottle (dimensions, approximately 50 mm high by 25 mm diameter) which may be tared before or after the tests. Dry the bottle and contents for 1 1/2 hours at a temperature ranging from 98° to 102°C, cool in a desiccator and weigh[1]. For standard operation the room temperature must be between 15° and 32°C. Wash the cup of the generating bulb free of any nitrogenous impurities and transfer the dried nitrocellulose to the cup. From a graduate or small beaker, transfer 25 ml of nitrogen-free 94.5 ± 0.5 percent sulfuric acid in small portions to the weighing bottle and then to the generating bulb cup until a sufficient quantity has been used to make an emulsion with the nitrocellulose. Lower the mercury reservoir sufficiently to cause reduced pressure in the generating bulb. Open wide the upper stopcock and open the lower stopcock sufficiently to draw the nitrocellulose into the generating bulb. Make successive rinsings with sulfuric acid until all of the nitrocellulose has been drawn in with the 25 ml of acid. It will generally be found possible to manipulate the upper stopcock after the first time. Force out any air drawn into the bulb, close the upper stopcock while

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[1]An alternate method of drying a sample for the determination of nitrogen content is as follows: Approximately 5 gm of the sample are dried in a moisture teller at 70°C for approximately 10 minutes. Approximately 1 gm is then transferred to a ground-glass stoppered weighing bottle and dried for 45 minutes at a temperature of 98° to 102°C.

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leaving the lower one open, and adjust the mercury reservoir so that the level of the mercury in the reservoir is from 13 to 15 inches below the level of the mercury in the generating bulb. Shake the generating bulb gently keeping the lower end in a fixed position until most of the gas has been generated. Adjust the mercury reservoir so the mercury in the generating bulb drops nearly to the lower shoulder, then close the lower stopcock and shake the bulb vigorously for 3 minutes. Replace the bulb on the rack, open the lower stopcock, and adjust the mercury reservoir until the mercury in the generating bulb is at approximately the same height as the mercury in the reservoir. This will bring the pressure inside the generating bulb to approximately 1 atmosphere, and the solubility of the nitric oxide gas in the sulfuric acid will be normal. Close the lower stopcock and shake the bulb vigorously for an additional 3 minutes. Transfer the gas to the measuring tube and adjust the levels of the mercury in the compensating and measuring tubes so that they are approximately the same and the level of the mercury in the compensating tube is approximately at the standardization mark. Allow the gas to stand for approximately 20 minutes in order to permit equalization of the temperature of the gas in the 2 tubes. Adjust the levels of the mercury in the tubes so that they are the same and that the level of the mercury in the compensating tube is at the standardization mark. Use a leveling device which will make possible the accurate adjustment of the levels. Divide the reading of the measuring tube by the weight of the sample of nitrocellulose to obtain the percentage of nitrogen in the nitrocellulose.

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Method No. 209.4

NITROCELLULOSE, ETHER ALCOHOL SOLUBILITY

1. PROCEDURE

1.1 Place two grams of the dry nitrocellulose obtained from the 20 gram stock in a 200 ml graduated cylinder, add 67 ml alcohol, 133 ml ether, stopper and shake well at frequent intervals for ten hours. Settle and pipette off 100 ml of clear solution into a tared beaker. Make a water precipitation as usual, using 50 ml of water. Evaporate and dry at 100°C. Calculate percent of soluble nitrocellulose. When more exact procedure is necessary use the following method:

1.2 Ether-alcohol solubility.--The official temperature for this determination shall be 15.5°C and in any case where the material does not conform to the requirements of the specification or where greater accuracy is desired, the complete determination shall be made at this temperature, bringing the solvents to this temperature before their addition to the sample. Use ether and alcohol conforming to the requirements of the applicable specification.

1.2.1 Volumetric method.--Place 1 gm of the dry sample in an Erlenmeyer flask, add 75 ml of alcohol, shake thoroughly for approximately 5 minutes, and allow the mixture to stand for 2 hours. Add 150 ml of ether, shake the mixture thoroughly, and allow it to stand overnight without removing the stopper of the flask. It is desirable to start this determination early in the day in order to give it plenty of agitation before being allowed to stand overnight, as agitation facilitates solution. After standing overnight, shake the mixture in the flask thoroughly and transfer it to a special solubility tube having a total length of 21 inches, inside diameter of 1.35 inches, and having the lower end constricted for length of 2 inches to an inside diameter of 0.3 inch and graduated in divisions of 0.1 ml. The constricted portion should be tapered gradually from the large part of the tube. Allow the mixture to remain in the solubility tube until the insoluble material has settled completely as indicated by the volume of this material remaining constant over a period of 1 hour. In the case of Grade A nitrocellulose determine the volume of insoluble and semisoluble material from the scale on the tube. If the volume is 0.25 ml or less, consider the percentage of insoluble material to be 1 percent or less. Should the volume exceed 0.25 ml use the appropriate gravimetric method.

1.2.2 Gravimetric method.--

1.2.2.1 Grade A.--Siphon off the clear liquid in the solubility tube to approximately 1/4 inch from the upper surface of the settled material. Add alcohol to 1/3 the capacity of the tube, shake vigorously, add ether to within 3 inches of the top of the tube, shake vigorously, and allow the mixture to settle. Repeat this process of washing by decantation 4 times. After the fourth decantation, wash the insoluble matter into a beaker with a small amount of ether-alcohol mixture. Filter the contents of the beaker through a tared filtering crucible having an ignited asbestos mat, and wash the residue on the filter with ether-alcohol mixture until it is free from soluble nitrocellulose. Dry the crucible and contents at a low temperature until no odor of ether can be detected, dry at 100°C to constant weight, cool in a desiccator and weigh. Carefully ignite the crucible over a free

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flame, cool in a desiccator, and weigh. Consider the loss in weight during ignition as that of insoluble nitrocellulose, and calculate this as percentage of the sample. Subtract this value from 100 percent to obtain the percentage of the nitrocellulose soluble in ether-alcohol.

1.2.2.2 Grades B and C.--Draw off 100 ml of the clear solution from the solubility tube without disturbing the settled layer of insoluble material. Transfer the 100 ml of clear solution to an aluminum can 50 mm in height by 90 mm in diameter having a tightly fitting cover, which has previously been dried and weighed. Precipitate the soluble material by the addition of distilled water, so as to obtain a finely divided precipitate of fibrous appearance. In order to obtain this, first heat the solution on a steam hot plate, without permitting the aluminum can to come in contact with steam or hot water, so as to evaporate part of the ether. Add 50 ml of distilled water gradually, with continued heating and stirring. The exact procedure for obtaining the best results must be determined by experience. After the nitrocellulose has been precipitated, continue the heating until the liquid is evaporated just to dryness. Dry the can and contents at 95° to 100°C for 1 1/2 hours, cool in a desiccator and weigh. Dry for an additional 1/2 hours, cool, and weigh to insure drying to constant weight. Consider the increase in weight of the can to be that of soluble nitrocellulose and calculate the percentage as follows:

$$\text{Percentage of soluble nitrocellulose} = \frac{100A}{W}$$

where

A = weight of soluble nitrocellulose in 100 ml of clear solution
W = four-ninths of the weight of the original sample

1.2.2.3 Alternate method, Grades B and C.--Transfer 0.3 gm of dried nitrocellulose to a 150 ml beaker, add 45 ml alcohol-ether mixture (2 ether-1 alcohol by volume) and let stand overnight under a bell jar. Transfer to a 100 ml centrifuge tube using alcohol-ether mixture and making up the volume to the 100 ml mark. Centrifuge at 2,000 r.p.m. for 10 minutes. Decant off the supernatant liquid, and refill with alcohol-ether mixture. Shake thoroughly, and centrifuge as before. Repeat 2 more times, a total of 4 operations. Carefully pour off most of the liquid in the centrifuge tube, leaving just enough to aid in transfer of the insoluble matter to a prepared tared, filtering crucible. Filter, wash well with the standard alcohol-ether mixture, dry for 30 minutes at 100°C, cool and weigh. Calculate the weight of the residue as percentage of nitrocellulose/in soluble in ether-alcohol.

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Method No. 210.1

STARCH

1. PROCEDURE

1.1 Transfer a weighed portion of exactly 5 grams of the sample to a 500 ml flask. Add 200 ml of distilled water and 5 ml of 70 percent nitric acid, and heat on a steam bath for three hours to hydrolyze the starch. To insure complete hydrolysis, break up or macerate the granules with a stirring rod having a flattened end. Filter the mixture through a filtering crucible and wash the residue with 200 ml of hot distilled water by filling the crucible, allowing the water to run through, and repeating the process. Transfer the combined filtrate and washings to a 600 ml beaker and evaporate to a volume of not less than 10 ml, carrying out the last part of the evaporation on a steam bath to prevent overheating of the residue. Cool the beaker and contents and add 50 ml of 40 percent sulfuric acid and 1.250 grams of powdered C.P. potassium dichromate. Cover the beaker with a watch glass, heat on a hot plate, and boil gently for 30 minutes. Cool the solution, transfer to a flask, and dilute to 200 ml with distilled water. Add 50 ml of a 10 percent solution of potassium iodide and titrate the liberated iodine with approximately 0.6N thiosulfate solution, using starch indicator near the end point. Calculate the percentage of starch in the sample on a moisture free basis.

$$\text{Percent starch} = \frac{(1.250 - VNF)0.137 \times 100}{W}$$

where

V = ml of sodium thiosulfate used
N = normality of sodium thiosulfate solution
F = milliequivalent weight of potassium dichromate = 0.04904
0.137 = grams of starch equivalent to one gram potassium dichromate
W = weight of sample corrected for moisture

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AGAR AGAR

Method to be decided.

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Method No. 212.1

AURINE

1. PROCEDURE

1.1 Add a five gram sample of (E.C.) Powder to a 200 mm drying tube which has the bulb packed with glass wool. Pack the glass wool tightly enough to prevent powder grains from passing through but not so tightly that the filtering proceeds too slowly. Extract the aurine with 10 - 15 portions of freshly prepared 1 to 1 acetone-water; wait for each portion to filter through before adding the next. Stir the sample to facilitate complete extraction by preventing channeling and filter directly into a 250 ml volumetric flask. Remove the flask and add the acetone-water solution until the flask is filled up to the mark. Stopper and shake well. Pour a sample into the colorimeter tube and measure the scale deflection. Read the percent aurine in the sample from the calibration curve of the colorimeter.

1.2 Colorimeter Calibration Curve.--Using fresh 1 to 1 acetone-water solution, prepare a series of 250 ml volumetric solutions containing the following quantities of aurine: 0.005, 0.010, 0.015, 0.020 and 0.025 grams. Fill the colorimeter tube with each solution and measure the scale deflection. Prepare a calibration curve, plotting percent aurine against scale reading. Make the readings within two hours after extraction. Note: Aurine solutions are stable for at least 24 hours if kept out of sunlight. Standardize the colorimeter weekly to avoid instrumental errors.

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Method No. 213.1

NITROGUANIDINE

1. PROCEDURE

1.1 Following the extraction with CCl_4 (Method No. 105.1), dry the residue in the thimbles to constant weight at 100°C. Weigh the thimbles every two hours during the drying and consider the weight as constant when the loss from the previous weight does not exceed 3 mg. Transfer the dried residue to a 400 ml beaker and boil for 30 minutes with each of four 150 ml portions of distilled water, decanting the boiling solution after each extraction. Filter the residue using the original extraction thimble and transferring the residue from the beaker to the thimble by means of a stream of hot water. Dry the thimble and residue in an oven at 100° ± 2°C to constant weight.

$$\text{Percent nitroguanidine} = \frac{100\ (A - B)}{W}$$

where

A = weight of thimble and contents prior to water extraction
B = weight of thimble and contents after water extraction
W = weight of sample corrected for total volatiles

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KETONONE

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Method No. 215.1

GLYCOLLATE

No method available.

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Method No. 216.1

RICINOLEATE

No method available.

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GROUP 300

ANALYTICAL TESTS OF INORGANIC INGREDIENTS

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Method No. 301.1

POTASSIUM SULFATE (TITRATION METHOD)

1. PROCEDURE

1.1 Accurately weigh 2 g of the powder and transfer in small portions with shaking to a 500 ml glass-stoppered Erlenmeyer flask containing 250 ml of dry acetone. (The acetone is dried by shaking with anhydrous calcium chloride and filtering). Shake on a mechanical shaker until solution of the powder grains is complete except for the slight suspension of graphite and potassium sulfate. Add 50 ml of glacial acetic acid, mix well and let stand for four hours or preferably overnight. Filter through a Whatman No. 42 or equivalent filter paper using suction (a Buchner funnel is desirable). Rinse the flask and filter with three 10 ml portions of dry acetone. Wash the filter with two 25 ml portions of dry acetone. Discard filtrate and replace with a clean 250 ml flask. Wash original flask with three 10 ml portions of hot water, and pass through the filter. Wash residue on filter paper with 40 ml of hot water. Boil down to a volume of 25 ml. Cool to room temperature. Add one drop of one percent phenolphthalein solution and make just alkaline by the dropwise addition of approximately 0.02N NaOH. Just discharge the pink color with approximately 0.02N HCl. Add one dipperful (0.2 ml) of tetrahydroxyquinone indicator and shake until completely dissolved. Add 25 ml of 95 percent ethyl alcohol and mix. Titrate dropwise with standard 0.04N $BaCl_2$, shaking vigorously after each addition, until a permanent rose-red color is obtained. The $BaCl_2$ solution can be standardized against C.P. K_2SO_4 as above.

$$\text{Normality} = \frac{\text{g of } K_2SO_4}{(0.0871)\ (\text{ml of } BaCl_2)}$$

Calculate the percentages of K_2SO_4 in the sample.

$$\text{Percent } K_2SO_4 = \frac{(\text{ml of } BaCl_2)\ (\text{normality})\ (8.713)}{\text{Weight of sample}}$$

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POTASSIUM SULFATE (COBALTINITRITE METHOD)

1. PREPARATION OF REAGENT

1.1 Dissolve 100 gm of chemically pure trisodium cobaltinitrite $Na_3Co(NO_2)_6 \cdot H_2O$ in water and dilute the solution to 500 ml. Store the solution in a dark bottle and avoid direct sunlight. This reagent is stable for only one week. Reagents older than one week should be discarded, as they tend to give high results. Standardize each lot of trisodium cobaltinitrite against a known solution of K_2SO_4 containing 0.050 gm per 25 ml. In the standardization, use a 20-percent solution of trisodium cobaltinitrite. The trisodium cobaltinitrite used in the analysis shall be a blend representing each bottle in the lot. The recovery shall be within ±3 percent of the theoretical.

2. PROCEDURE

2.1 Weigh a 2-gm sample into a tared 90-mm evaporating dish, add 10 ml of 70-percent HNO_3 and place on the steam bath until the reaction starts. Turn off the steam until the reaction subsides, then evaporate to a gummy mass. (Do not take to dryness on a hot plate.) Place the dish on a nichrome triangle on a tripod and heat cautiously from the top by means of a burner. Finally heat from the bottom until the bulk of the carbonaceous matter is removed, then transfer to a muffle furnace at dull red heat for 10 to 15 minutes. A small amount of carbon can be neglected. Allow the dish to cool, rinse the sides with 5 ml of water, then add 5 ml of concentrated HCl. Cover the dish with a watch glass, place the dish on the steam bath for 5 to 10 minutes, dilute the contents of the dish to approximately 25 ml, filter the solutions through a filter paper into a 150 ml beaker, and wash the residues thoroughly. Evaporate the filtrate to dryness on the steam bath. Add 25 ml of 1 N HNO_3, and cool to 20°C. By means of a graduate, add 15 ml of cooled (20°C) 1 N trisodium cobaltinitrite solution with stirring, to the cooled solution containing potassium ions. Cover the resulting mixture with a watch glass and allow to stand for 2 hours at 20°C. Filter the solution through a tared filtering crucible. Use a minimum amount of 0.01 normal nitric acid solution in a wash bottle to transfer the precipitate of dipotassium-sodium cobaltinitrite from the beaker to the crucible. Wash the precipitate with three 10-ml portions of 0.01 normal nitric acid solution and then with three 5-ml portions of alcohol. Aspirate the crucible until the contents are fairly dry, place the crucible and contents in an oven at 100°C for 1 hour, cool in a desiccator and weigh.

$$\text{Percent potassium sulfate} = \frac{38.4A}{W}$$

where

A = gm of precipitate
W = gm of the volatiles free sample

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Method No. 301.3

POTASSIUM SULFATE (IGNITION METHOD)

1. PROCEDURE

1.1 Transfer to a previously ignited and cooled 50-ml porcelain crucible an accurately weighed portion of approximately 2 g of the sample. Add 5 ml of concentrated nitric acid and place the crucible on a steam bath. Allow the digestion to continue until the powder is completely disintegrated and no liquid remains. Dissolve the contents of the crucible in about 100 ml of hot distilled water in 20 ml portions and filter the solution, catching the filtrate in a 250-ml beaker. Render the solution slightly acid with HCl and heat to boiling. Add slowly 5 ml of a 10-percent solution of barium chloride and continue the boiling for several minutes. Cover the beaker with a watch-glass, place on a steam bath, and allow the precipitated barium sulfate to settle. Catch the precipitate on a tared Gooch crucible and wash well with hot distilled water. Ignite the crucible and contents, cool in a desiccator, and weigh. Calculate the weight of precipitate to percentage of potassium sulfate in the sample as received.

$$\text{Percentage of potassium sulfate} = \frac{(\text{Wt. ppt.} \times 74.65)}{\text{Wt. sample}}$$

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POTASSIUM NITRATE

1. PROCEDURE

1.1 Weigh 5 g of the original sample into a 90 mm evaporating dish, add 10 ml of 70 percent HNO_3 and place on a steam hot plate in a hood. After the reactions commence, remove from the hot plate until the violent evolution of NO_2 fumes slow down, then evaporate to a gummy mass. Place the dish on a triangle and heat over a low flame, beginning at one side of the dish. Gradually increase the heat until the carbonaceous matter is nearly burned off. After the dish has cooled, rinse the sides of the dish with 5 ml of water, then carefully add 5 ml of concentrated HCl. Cover with a watch glass and place on the steam bath for five to ten minutes. Dilute the contents of the dish to about 25 ml and filter through a filter paper into a 150 ml beaker. Add 10 ml of 5 percent H_2SO_4, and evaporate on a hot plate until most of the excess H_2SO_4 has been driven off. Place the beaker on a tripod and heat with a flame until all H_2SO_4 has been evaporated. Add 20 ml of water, cool to 20°C and add 5 ml of N/1 HNO_3. By means of a graduate, add 15 ml of cooled (20°C) $Na_3Co(NO_2)_6$ reagent, with stirring. Let stand for two hours at 20°C and filter through a tared Gooch crucible, washing the precipitate first with three 10 ml portions of N/100 HNO_3 then with three 5 ml portions of alcohol. Dry at 100°C for one hour, cool in a desiccator and weigh.

$$\frac{\text{Weight of } K_2NaCo(NO_2)_6H_2O \times 44.52}{\text{Weight of sample corrected for M. \& V.}} = \text{Percent Potassium nitrate}$$

Note: (1) Dissolve 200 g of chemically pure trisodium cobaltinitrate $Na_3Co(NO_2)6.H_2O$ in water and dilute the solution to 500 ml. Store the solution in a dark bottle and avoid direct sunlight. This reagent is stable for only one week. Reagents older than one week should be discarded, as they tend to give high results.

(2) If the potassium nitrate content of the powder exceeds one percent, use a smaller sample or take an aliquot portion of the solution after dissolving the ash in hydrochloric acid.

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Method No. 302.2

POTASSIUM NITRATE

1. PREPARATION OF REAGENT

1.1 Dissolve 200 g of chemically pure trisodium cobaltinitrite ($Na_3Co(NO_2)_6.(H_2O)$) in water and dilute the solution to 500 ml. Store the solution in a dark bottle and avoid direct sunlight. This reagent is stable for only 1 week. Reagents older than 1 week should be discarded, as they tend to give high results.

2. PROCEDURE

2.1 Transfer the solution containing the potassium salt remaining from the determination of barium nitrate given in Method No. 304.1, (or an aliquot portion thereof, if the amount of potassium nitrate in the powder exceeds 1 percent) to a 150 ml beaker and evaporate the solution on a hot plate in a hood until the evolution of SO_3 fumes ceases. Cool the beaker and contents, add 25 ml of water and warm the solution on a steam bath with stirring. If no insoluble material is present, add 5 ml of approximately normal HNO_3 and cool the solution to 20°C. If the solution warmed on the steam bath was found to contain insoluble material, filter the solution through filter paper and wash with hot water. Evaporate the filtrate and washings to dryness on the steam bath, add 25 ml of water and 5 ml of approximately normal HNO_3, and cool the solution to 20°C. By means of a graduate, add 15 ml of cooled (20°C) 2-normal trisodium cobaltinitrite solution with stirring, to the cooled solution containing potassium ions. Cover the resulting mixture with a watch glass and allow to stand for 2 hours at 20°C. Filter the solution through a tared filtering crucible containing a mat of medium thickness. Use a minimum amount of 0.01 normal HNO_3 in a wash bottle to transfer the precipitate of dipotassium-sodium cobaltinitrite from the beaker to the crucible. Wash the precipitate with three 10 ml portions of 0.01N HNO_3 and then with three 5 ml portions of alcohol. Dry the crucible and contents in an oven at 100°C for 1 hour, cool in a desiccator and weigh. Calculate the increase in weight of the crucible to percentage of potassium nitrate as follows:

$$\text{Percent potassium nitrate} = \frac{44.52A}{W}$$

where

A = weight of precipitate
W = weight of the volatiles-free sample

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Method No. 303.1

POTASSIUM PERCHLORATE

1. APPARATUS AND REAGENTS

1.1 An Emerson or a Parr Calorimetric Bomb having a capacity of 350 to 400 ml may be used.

1.2 0.05-normal silver nitrate solution.--Dissolve 16.989 grams of reagent grade silver nitrate of known purity in distilled water, transfer quantitatively to a 2-liter volumetric flask and make up to volume.

1.3 0.05-normal potassium thiocyanate solution.--Dissolve 4.860 grams of C.P. grade potassium thiocyanate in distilled water, transfer to a one liter volumetric flask and make up to volume. Determine by a blank titration the ml of the 0.05-normal silver nitrate solution equivalent to 1 ml of this solution.

1.4 Ferric ammonium sulfate indicator.--Dissolve 200 grams of C.P. grade ferric ammonium sulfate in approximately 300 ml of water. Acidify the solution with 25 ml of approximately 70 percent nitric acid solution.

2. PROCEDURE

2.1 Transfer an accurately weighed 5-gram representative portion of the powder to the bomb. Connect the fuse wire to the electrodes and arrange it to make close contact with the powder. Seal the bomb assembly securely. Ignite the powder by passing through the fuse wire an electric current which has been found to cause the wire to glow for several seconds before it disintegrates. Cool the bomb. Slowly release the gas pressure and disassemble the bomb. Wash the contents of the bomb completely into a beaker with small portions of water, making sure that all fittings are thoroughly rinsed. Filter the combined washings and wash the filter with water, catching the filtrate in a beaker. Evaporate the filtrate to a volume of approximately 100 ml, allow to cool, transfer to a 250 ml volumetric flask and make up to volume.

2.2 Pipette a 25 ml aliquot of the solution into a 250 ml iodine flask. Add 10 ml of approximately 35 percent nitric acid solution, 3 to 5 ml of the ferric alum indicator solution, 1 to 2 ml of nitrobenzene and 10.00 of 0.05-normal silver nitrate solution, which should be in excess. Shake vigorously for several seconds to coagulate the silver chloride in the nitrobenzene. Titrate the excess silver nitrate to the first permanent coloration with 0.05-normal potassium thiocyanate. Calculate the potassium perchlorate content of the sample as follows:

$$\text{Percentage of potassium perchlorate} = \frac{(A-BC)\ N\ 138.6}{W}$$

where

A = ml of silver nitrate solution added
B = ml of potassium thiocyanate solution added
C = ml of silver nitrate solution equivalent to 1 ml of potassium thiocyanate solution
N = normality of silver nitrate solution
W = weight of sample, grams

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Method No. 304.1

BARIUM NITRATE

1. PROCEDURE

1.1 Transfer a weighed portion of approximately 5 gm of powder sample to a 90-mm porcelain evaporating dish, add 15 ml of approximately 70-percent HNO_3 solution, and place the dish and contents on a steam bath in a hood. Heat until reaction becomes evident, remove the dish and contents from the steam bath, and allow the reaction to proceed until evolution of fumes has nearly ceased. Replace the dish and contents on the steam bath and heat the mixture until only a gummy residue remains. Place the dish and contents on a nichrome triangle on a tripod and heat with a low flame directed only on the sides of the evaporating dish until the carbonaceous matter is nearly all burned off. High temperatures should be avoided, since they result in fusion of the salts with the porcelain. Allow the dish and contents to cool, rinse the sides of the dish with 5 ml of water, and then with 5 ml of approximately 38-percent HCl. Cover the dish with a watch glass, place the dish on the steam bath for 5 to 10 minutes, dilute the contents of the dish to approximately 25 ml, filter the solution through a filter paper into a 150-ml beaker, and wash the residue thoroughly. Dilute the filtrate and washings to approximately 75 ml, heat the solution to boiling and add 10 ml of 5-percent H_2SO_4 with stirring. Digest the mixture on the steam bath for at least 1 hour, and filter the solution through a previously ignited and tared filtering crucible. Retain the filtrate for the determination of potassium nitrate. Dry the crucible and contents in an oven at 100°C for approximately 1 hour, ignite at dull red heat in a muffle furnace for 15 to 30 minutes, cool in a desiccator, and weigh. Calculate the increase in weight to percentage of barium nitrate as follows:

$$\text{Percent barium nitrate} = \frac{112.0\ A}{W}$$

where

A = weight of precipitate
W = weight of volatiles-free sample.

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Method No. 305.1

CALCIUM CARBONATE-SODIUM SULFATE (W. C. C. METHOD)

1. PROCEDURE

1.1 Weigh five grams (plus or minus 0.001g) of powder and transfer to a clean dry 250 ml beaker. Add 50 ml of morpholine to the beaker and heat on a hot plate under the hood until the powder is dissolved. CAUTION: Morpholine fumes are inflammable. If fumes are seen to be going over the top of beaker, remove from hot plate and allow to cool slightly. Then replace and continue heating until the powder is dissolved. Filter the hot solution through a tared Gooch crucible (A), pouring it down a stirring rod. After the transfer, place 25 ml of morpholine in the beaker and heat until white fumes appear. While it is still fuming, pour the morpholine down the stirring rod so that the hot liquid comes in contact with and washes down anything that may have been left on the sides of the crucible from the first transfer. Remove the hot morpholine from the filtering flask and cool the flask by rinsing it out with cold water. Pour 25 ml of anhydrous ether into the beaker and with the aid of a policeman transfer all of the residue from the beaker into the Gooch crucible. With a clean towel, wipe off the outside of the crucible, then place in a 100°C oven for 30 minutes. Remove, cool in a desiccator, and weigh (B). Wash the residue in the crucible with four 25 ml portions of warm distilled water. Pour a small amount of ether through the crucible and again dry 30 minutes at 100°C, after it is free from ether fumes. Remove from the oven, cool in a desiccator for 20 minutes and reweigh (C).

2. CALCULATIONS

2.1 Calculation for percent Na_2SO_4:

$$\frac{\text{Loss in weight (B-C) x 100}}{\text{Wt. of sample}} = \text{percent } Na_2SO_4$$

Wash the residue in the crucible with 50 ml of dilute (3N) HCl. Wash with 50 ml of distilled water and finally with 25 ml of anhydrous ether. Dry for 30 minutes at 100°C. Remove from oven, cool in a desiccator and reweigh (D).

2.2 Calculation for percent $CaCO_3$:

$$\frac{\text{Second loss in weight (C-D) x 100}}{\text{Wt. of sample}} = \text{percent } CaCO_3$$

2.3 Calculation for percent Graphite:

$$\frac{\text{Final weight of crucible - tare (D-A) x 100}}{\text{Wt. of sample}} = \text{percent Graphite}$$

If the powder has a high ash determine true graphite by heating the crucible in a muffle or over a Meker (or Fisher) burner. Cool and reweigh, the loss in weight = Graphite.

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Method No. 306.1

TIN (TITRATION)

1. PROCEDURE

1.1 Transfer an accurately weighed portion of approximately 2 gm of the powder to a 300-ml flask equipped with a side arm. Add 5 ml of concentrated H_2SO_4 and 5 ml of concentrated HNO_3. Heat on a hot plate to decompose the powder and continue heating to the evolution of SO_3 fumes. Cool, add 5 ml of concentrated HNO_3 and again heat to the evolution of SO_3. Repeat the addition of HNO_3 and heating until the organic matter is destroyed and the solution is clear and colorless after cooling with the exception of a little graphite. Add 2 ml of water and again heat to SO_3 fumes. Repeat the addition of 2 ml of water and heating to insure the complete removal of HNO_3. Cool. Add 100 ml of water, 10 ml of concentrated HCl and filter through ashless paper. Add 2 strips of nickel (approximately 1 square inch in area by 0.020 inch nominal thickness) and heat the solution at such a rate that a copious stream of fine bubbles rises from the strips, but no appreciable boiling occurs. Reduce the solution for 45 minutes. Cool the flask and contents, keeping the liquid under an atmosphere of CO_2 either by passing a fine stream of CO_2 into the flask or by means of a goose-neck, one end of which extends into a 10-percent $NaHCO_3$ solution. On cooling, some $NaHCO_3$ is drawn into the flask with the resultant evolution of CO_2. Cool the flask to approximately 20°C and titrate with 0.02 N iodine solution using starch solution indicator. Calculate the percent tin as follows:

$$\text{Percent tin} = \frac{5.935\ VN}{W}$$

where

V = ml iodine solution used in titration
N = normality of iodine solution used
W = gm of sample (volatiles-free)

Method No. 306.2

TIN (GRAVIMETRIC)

1. PROCEDURE

1.1 Weigh 2 grams of sample into a small porcelain casserole, add 25 ml of nitric acid (1.42 SpG) and allow to decompose. Evaporate to a low bulk, dilute with 1 percent nitric acid, warm and filter through a Gooch or paper, wash with warm water. Burn off in a crucible or dry the Gooch at 100°C and weigh. Report as Tin or Tin Oxide.

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CRYOLITE (SODIUM ALUMINUM FLUORIDE)

1. REAGENTS

1.1 Mix 80 g of uranyl acetate dihydrate, 48 g of 30 percent acetic acid, 390 ml of distilled water and warm to dissolve all salts. In another container, mix 240 g of zinc acetate trihydrate, 25 g of 30 percent acetic acid, 250 ml of distilled water and warm to dissolve all salts. Mix both solutions and allow to stand for at least 24 hours. The resulting reagent shall be designated as Reagent A. Filter before use as required since precipitation may occur as a result of sodium impurities. Store in a pyrex glass vessel.

1.2 Wash Liquid.--Saturate 95 percent ethyl alcohol with sodium zinc uranyl acetate, for wash liquid.

2. PROCEDURE

2.1 Mix a ground and weighed portion of approximately 5 g of the sample with 10 g of molten paraffin wax in a silica dish. Heat gently to ignition and allow to burn off. When the flame is extinguished, place the dish in a muffle furnace at a dull red heat until the residual carbon is burned off and then cool. Dissolve the ash in a few ml of aqua regia (3 parts concentrated nitric acid to 1 part concentrated hydrochloric acid), evaporate to dryness and redissolve the residue in approximately 5 ml of distilled water. At this point it is important that all reagents and solution be maintained at a steady temperature for the remainder of the estimation. Add 20 ml of Reagent A (filtered just before use), stir vigorously for 15 seconds with a thin glass rod. Allow to stand at room temperature for one hour and then filter through a tared filtering crucible. Wash five times with 2 ml portions of wash liquid and finally with ether. Draw air through the crucible for 15 minutes until the ether is removed and weigh. Calculate the percentage of cryolite as follows:

$$\text{Percent cryolite} = \frac{4.55A}{W}$$

where

A = grams of precipitate (sodium zinc uranyl acetate hexahydrate)

W = grams of sample

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Method No. 308.1

GRAPHITE

1. PROCEDURE

1.1 Transfer a weighed portion of approximately 10 gm of the sample to a 400-ml beaker, add 75 ml of approximately 70-percent HNO_3 heat the mixture on a steam bath. When the reaction starts, remove the beaker and allow the reaction to proceed to completion. The rate of decomposition can be controlled, if necessary, by adding a little distilled water from time to time. After evolution of oxides of nitrogen ceases, heat the material if all the particles have not been completely decomposed. Cool the beaker and contents, and filter the solution through a tared filtering crucible prepared with a thin layer of asbestos which has been washed with 10 to 15 ml of approximately 70-percent HNO_3 and then with water. Wash the residue with hot water, remove the filtrate which contains nitric acid, wash the residue with acetone until all the organic material is dissolved as shown by the absence of coloration of the washings. Wash again with water and finally with acetone. Dry the crucible and contents at 100°C for 1 hour, cool in a desiccator and weigh. Calculate the increase in weight of the crucible to percentage of graphite in the powder.

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CARBON BLACK

1. PROCEDURE

1.1 Transfer an accurately weighed 5-gram portion of the finely divided powder to a 400 ml beaker. Add 75 ml of approximately 70 percent nitric acid solution, cover the beaker with a watch glass and place the assembly on a steam bath until the reaction starts. Remove the beaker from the steam bath until the reaction subsides then replace the beaker on the bath and allow the mixture to digest for approximately one hour. Chill the mixture by immersing the beaker in an ice bath and add 75 ml of distilled water. Allow the beaker and contents to stand until the major part of the carbon has settled out and then filter the solution through a tared fritted glass, alundum or Gooch crucible. Wash the residue containing the carbon black with portions of hot water, until the washings are colorless, allowing each portion of water to be drawn through the crucible before adding the next. Dry the crucible in an oven maintained at 100° to 105°C for 20 minutes, cool, and then wash the residue with several portions of acetone (Federal Specification O-A-51a, Grade A or equivalent) until the washings are colorless. Dry the crucible at 200° to 205°C for 2 hours, cool in a desiccator for 30 minutes and weigh. Calculate the increase in weight of the crucible to percentage of carbon black in the powder.

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Method No. 309.2

CARBON BLACK (PHOTOMETRIC METHOD)

1. APPARATUS

1.1 Filter photometer or spectrophotometer with red-sensitive photocell.

1.2 Wratten filter No. 87 for filter photometer, or setting of between 800 and 900 millimicrons for the spectrophotometer.

1.3 Matched absorption cells with 1-cm light path. These may be of any type adapted to the photometer used.

1.4 The following solutions are prepared by dissolving the indicated amounts of powder in redistilled acetone and making the solutions up to 100 ml in volumetric flasks.

A. 5.00 gm Reference Powder A. (See notes 1 and 2 for compositions of Reference Powders A and B.)
B. 0.300 gm Reference Powder B and 4.70 gm Reference Powder A.
C. 0.400 gm Reference Powder B and 4.60 gm Reference Powder A.
D. 0.500 gm Reference Powder B and 4.50 gm Reference Powder A.
E. 0.600 gm Reference Powder B and 4.40 gm Reference Powder A.
F. 5.00 gm powder to be analyzed.

A sufficient quantity of each of these solutions to fill the absorption cells is centrifuged at 3000 rev/min for 15 minutes to remove potassium sulfate and dirt. The absorption cells are then filled, stoppered or otherwise covered to prevent evaporation, and the transmission measured as described in paragraphs 2 and 3 below.

2. PREPARATION OF WORKING CURVE

2.1 Place the cell containing solution A in the photometer and adjust the instrument to read 100 percent transmission. This cell is to be used as a reference in all subsequent measurements.

2.2 Place the cell containing solution B in the photometer and measure the transmission of this solution.

2.3 Recheck the adjustment of the instrument with the cell containing Solution A; if the reading is less than 99 percent, readjust the instrument and repeat.

2.4 Repeat steps 2 and 3 with solutions C, D, and E. The measurements should be repeated at least three times and the average transmission computed for each solution.

2.5 Plot the logarithm of the transmission against the carbon black concentration, expressed as percentage carbon black in the powder, for each solution. The percentage carbon black in the powder is computed from the formula.

$$\text{Percentage carbon black} = \frac{W_B}{5.00} \times C_B$$

where

W_B = weight of Reference Powder B in solution, and
C_B = percentage carbon black in Reference Powder B.

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3. DETERMINATION OF CARBON BLACK IN THE UNKNOWN POWDER

3.1 The photometer is adjusted to read 100 percent transmission with Solution A in place.

3.2 The cell containing solution F is placed in the photometer and the transmission measured.

3.3 The cell containing solution D is placed in the photometer and the transmission measured.

3.4 The transmission of solution A is rechecked; if it is less than 99 percent, steps 3 and 4 are repeated.

3.5 The percentage of carbon black in the unknown powder is read from the graph. Solution D is used as a check on the behaviour of the system. Its transmission should be within one percent of that previously measured.

3.6 A working curve should be prepared before each series of analyses (at least daily) and the check on solution D which should have very nearly the nominal carbon black content of the powder being analyzed, should be run as a check with each unknown.

4. NOTES

4.1 Composition of Reference Powder A.

Nitrocellulose (13.15% N)	57.50%
Nitroglycerin	39.25%
Ethyl Centralite	1.75%
Potassium Sulfate	1.50%

4.2 Composition of Reference Powder B.

Nitrocellulose (13.15%)	57.50%
Nitroglycerin	39.25%
Ethyl Centralite	1.75%
Potassium sulfate	1.50%
Added:	
Carbon Black (Raven #15 or equal)	0.50%

4.3 The centrifuge tubes must be stoppered to avoid changes in concentration due to evaporation.

4.4 It should not be necessary to determine correction factors for each cell unless the cell thicknesses vary by more than 0.5 percent.

4.5 The expected overall accuracy of the determination is $\pm$ 5 percent of the amount of carbon black present.

4.6 Best results will be obtained when the transmissions of the solutions are between 35 percent and 65 percent.

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GROUP 400

STABILITY TESTS

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Method No. 401.1

134.5°C HEAT TEST

1. PROCEDURE

1.1 Make tests on five samples thus: Weigh out 2.5 grams of sample and place each in a thoroughly cleaned and dry test tube made of heavy glass, preferably pyrex, approximately 15 millimeters inside diameter, 18 millimeters outside diameter, and 290 millimeters long. Each sample shall consist of as nearly whole grains as is consistent with a weight of 2.5 grams of sample. In sectioning grains to obtain the required weight, split the grains longitudinally. Place a piece of standard normal methyl violet paper, 70 millimeters long and 20 millimeters wide, vertically in each tube, its lower edge 25 millimeters above the powder. Stopper the tubes with corks through which holes 4 millimeters in diameter have been bored or which have been notched to a depth of 4 mm. Place the tubes in a constant temperature bath at 134.5 ± 0.5°C. (See U.S. Army Ordnance Department Drawing CP-24114 for apparatus used at Army activity) so that no more than 6 or 7 millimeters of length projects. Examine each tube by withdrawing until the top of the paper can be seen distinctly and replacing quickly at 5-minute intervals beginning 5 minutes before the minimum time stated for the specified class of powder to change the paper to salmon pink. Record as the time of completion of this part of the test, the time of the observation, in multiples of 5 minutes, which reveals the test paper in any tube to be completely changed to salmon-pink. This occurs when the paper is completely free of violet color. Continue heating and report whether any sample of single-base powder explodes in less than 5 hours, or EC powder in less than 4 hours. (Secure standard normal methyl violet paper and standards for salmon pink from the Ordnance Department.)

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Method No. 402.1

120°C HEAT TEST

1. PROCEDURE

1.1 Make tests on five samples thus: Weigh out 2.5 grams of sample and place each in a thoroughly cleaned and dry test tube made of heavy glass, preferably pyrex, approximately 15 millimeters inside diameter, 18 millimeters outside diameter, and 290 millimeters long. Each sample shall consist of as nearly whole grains/as is consistent with a weight of 2.5 grams of sample. In sectioning grains to obtain the required weight, split the grains longitudinally. Place a piece of standard normal methyl violet paper, 70 millimeters long and 20 millimeters wide, vertically in each tube, its lower edge 25 millimeters above the powder. Stopper the tubes with corks through which holes 4 millimeters in diameter have been bored or which have been notched to a depth of 4 mm. Place the tubes in a constant temperature bath at 120 ± 0.5°C. (See U.S. Army Ordnance Department Drawing CP-24114 for apparatus used at Army activity) so that no more than 6 or 7 millimeters of length projects. Examine each tube by withdrawing until the top of the paper can be seen distinctly and replacing quickly at 5-minute intervals beginning 5 minutes before the minimum time stated for the specified class of powder to change the paper to salmon pink. Record as the time of completion of this part of the test, the time of the observation, in multiples of 5 minutes, which reveals the test paper in any tube to be completely changed to salmon-pink. This occurs when the paper is completely free of violet color. Continue heating and at the end of the time intervals stated for the specified class of powder, remove the tubes, place in front of a white paper, and note and report if any fumes are present in any tube. (Secure standard normal methyl violet paper and standards for salmon pink from the Ordnance Department.)

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GROUP 500

PHYSICAL TESTS

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Method No. 501.1

DUST AND FOREIGN MATERIAL

1. APPARATUS

1.1 Using the apparatus, shown on sketch "Apparatus for the Determination of the Dust Content of Smokeless Rifle Powder" (Figure 13) which shall be maintained in good condition, the determination of dust shall be according to the following procedure:

2. PROCEDURE

2.1 Precondition the U-tube containing glass wool by blowing compressed air through it for a few minutes. Weigh to the nearest 0.1 g two 100-g powder samples. Weigh the U-tube to the nearest milligram and place it in position for the run. With the help of a small funnel, transfer one of the samples to the charging can; close the rubber hose with a pinch clamp, and return the can to its correct position. The nozzle must be placed not more than 0.08 inch from the opening in the T-tube beneath the charging cone.

2.2 After applying compressed air at approximately 19 pounds pressure, open the pinch clamp until the powder flows at such a speed that the sample requires about two minutes for passage. A small lead hammer is useful in rapping the apparatus to dislodge any clinging dust particles.

2.3 Replace the sample in the charging can, and pass it though the apparatus again. Repeat this procedure for a total of five passes.

2.4 Remove the U-tube and weigh to the nearest milligram. The increase in weight is equal to the percent dust content of the sample.

2.5 Replace the U-tube on the apparatus and run a duplicate determination with the other sample. Report the average of the two determinations to the nearest 0.001 percent.

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Method No. 501.2

DUST (ALTERNATE METHOD)

1. SAMPLING

1.1 A separate one-quart sample should be obtained directly from one of the powder boxes and reserved for the dust determination in order to prevent loss or settling of dust during analysis. The test should be run in duplicate and the results averaged.

2. APPARATUS

2.1 Use the assembly, shown in Figure 14, consisting of two 750 ml Erlenmeyer flasks (A and A') for holding duplicate samples, a 500 ml bottle (B) filled with glass wool to filter the air and a 1 liter suction flask (D) used as a pressure regulator.

3. PROCEDURE

3.1 Weigh out a 50-gram sample to 0.001 g and pour into 200 cc distilled water contained in a clean 750 cc Erlenmeyer flask. Care should be exercised in handling or pouring the powder to prevent loss or alteration in dust content. After inserting stopper in flask (A) with air inlet tube from 1/8 inch to 1/4 inch from bottom (see Figure 14), adjust air flow so that the liquid in the pressure indicator (E) is brought to a mark 24 inches above liquid level; allow to agitate for ten minutes. Shut off air, remove air inlet tube - with rinsing - disperse the dust by a swirling motion, and quickly decant into a one-liter beaker. Add 150 cc of water to the powder and repeat the swirling and decantation. Repeat until no more dust particles are noticeable. This should take about 650 to 800 cc total. Filter the dust suspension through a tared No. 3010 Selas crucible, dry for one hour at 100°C, cool in a desiccator and weigh.

$$\frac{\text{Weight of Residue} \times 100}{\text{Sample Weight}} = \text{Percent Dust}$$

To clean and condition crucibles, soak in dichromate H_2SO_4 cleaning mixture for one-half hour or more, rinse, and wash with distilled water, placing crucible in an inverted position on suction filter.

Method No. 501.3

DUST

1. PROCEDURE

1.1 Place approximately 100 g of the powder in a tared glass or aluminum dish, dry in a desiccator over 95 to 96 percent sulfuric acid for 18 to 20 hours, and then weigh. Pass the powder through a separator (Figure 15) in which the dust is separated from the powder grains by means of a stream of air of suitable velocity. Weigh the powder and again pass through the separator; the operation shall be repeated until constant weight is attained. Calculate the loss in weight of the powder as percentage of dust and foreign matter.

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Method No. 502.1

GRAVIMETRIC DENSITY

1. PROCEDURE

1.1 The apparent specific gravity is the weight of a unit volume as related to the weight of the same unit volume filled with water at a temperature of 70°F. For convenience use a du Pont Densimeter. The cup shall have a volume of 324.9 ml. The usual opening shall be 3/8 inch diameter, but 1/2 inch may also be used and must be so stated in report of analysis. Close valve, fill hopper level with 350 - 400 grams powder. Open valve quickly and completely, allow powder to flow until cup just overflows and then close valve. Remove hopper, run sharp straight edge across top of cup to remove excess, then tap cup to settle powder, replace on balance of densimeter and adjust weight to balance. Scale beam reads direct to give gravimetric density. For temperature above 70°F add 0.001 to gravimetric density for every two degrees, and for temperatures below 70°F subtract 0.001 from the gravimetric density for every two degrees. Maintain the powder for 24-36 hours at the temperature of the test before testing. The temperature range is critical in this determination.

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Method No. 503.1

HYGROSCOPICITY (EQUILIBRIUM METHOD)

1. PROCEDURE

1.1 Spread an accurately weighed sample of 10 to 15 g of the powder on a 3-1/2 inch watchglass and place in a desiccator containing sulfuric acid-water solution of such concentration as to give 20 percent humidity (59.10 percent H_2SO_4). Place the desiccator in an oven regulated at a constant temperature of 30°C. Weigh the powder every other day until a constant weight is obtained. This usually requires from six to seven days. After the sample has reached constant weight, at 20 percent humidity and 30°C, place it in a desiccator containing sulfuric acid of such concentration as to give 90 percent humidity (18.60 percent H_2SO_4), and place the desiccator in the constant temperature oven at 30°C, until the weight becomes constant. The hygroscopicity is the percent of moisture absorbed between the conditions of exposure at 20 percent and 90 percent relative humidity. Calculate the hygroscopicity as follows:

$$\text{Percent hygroscopic moisture} = \frac{(C-B) \times 100}{W}$$

where

W = weight of sample
B = constant weight at 20 percent humidity
C = constant weight at 90 percent humidity

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Method No. 503.2

HYGROSCOPICITY (TOTAL MOISTURE)

1. PROCEDURE

1.1 The test shall be made in duplicate. Transfer a sample of approximately 100 gm weight to a tared glass weighing bottle with a ground glass cover. Place the weighing bottle (without cover) in a humidor containing one liter of sulfuric acid-water mixture (18.6 $\pm$ 0.5 percent sulfuric acid) which will maintain a relative humidity of 90 $\pm$ 0.25 percent at 30°C. The humidor shall have a total capacity of about 10 liters and shall be such that it may be sealed hermetically. A 10-inch desiccator meets these requirements. Place the humidor in an oven maintained at 30° $\pm$ 2°C. On the fourth day and every day thereafter remove the glass weighing bottle, cover with its ground glass cover, cool in a desiccator and weigh. Repeat the weighings until the sample of powder ceases to gain in weight, indicating that the powder has reached equilibrium with 90 percent relative humidity. Determine the total moisture content of the powder by the method described under the paragraph for total moisture in the referenced detail specification using 100 $\pm$ 0.1 gm of the powder.

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Method No. 504.1

DIMENSIONS OF GRAINS

1. PROCEDURE

1.1 Length.--Select at random 30 normal grains of powder and measure the length with a micrometer. When the ends are irregular, determine the average length as closely as possible.

1.2 Grain diameter, perforation diameter, and web thickness.--Select 30 normal grains of either single or multi-perforated powder. By means of a razor blade or hacksaw, cut each grain partly through from opposite sides at about the middle, and break under strain so that a narrow ridge of powder which includes the desired line of measurement is left along the line of breakage. This desired line of measurement is a straight line forming a diameter of the grain which passes through the center of the perforation of a single-perforated powder and through the centers of three perforations of a multi-perforated powder. Place a number of the grains vertically on a glass slide so that the broken ends are uppermost and the individual desired lines of measurement are in alignment. Use a film of vaseline on the glass slide to hold the grains in position if necessary. Place the slide on the stage of a measuring microscope having a magnification of approximately 25 diameters, equipped with cross-hairs, and having a horizontal scale and vernier screw scale capable of giving measurements accurate to 0.001 inch. In the case of multi-perforated grains, measure the diameter of the grain, the two outer web thicknesses, the two inner web thicknesses, and the diameters of three perforations of each of the 30 grains by means of a continuous series of readings taken along the desired line of measurement of each grain. In the case of single-perforated grains, measure the diameter of the grain, the two web thicknesses and the diameter of the perforation of each of the 30 grains by means of a continuous series of readings taken along the desired line of measurement of each grain.

1.3 Length-diameter ratio.--From the measurements of length and grain diameter taken as described above, calculate the ratio between the average length (L) and the average diameter (D).

1.4 Length and diameter uniformity.--From the measurements of length and grain diameter taken as described above, calculate the mean variation of the individual dimensions from the mean dimension, expressed as percentage of the mean dimension.

1.5 Grain diameter - perforation diameter ratio.--From the measurement taken, calculate the ratio between the average grain diameter (D) and the average perforation diameter (d).

1.6 Web measurements.--

1.6.1 Type I. From the measurements made, calculate the average outer web thickness (Wo) and the average inner web thickness (Wi) for the 30 grains measured. Calculate the average web thickness (Wa) by taking the average of Wo and Wi. Calculate the difference between the average outer web thickness (Wo) and the average inner web thickness (Wi) in terms of percentage of the average thickness (Wa):

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$$\frac{(Wo-Wi) \times 100}{Wa} \text{ or } \frac{(Wi-Wo) \times 100}{Wa} = \text{percentage difference}$$

1.6.2 Type II. From the measurements made calculate the average of the 30 web thicknesses above the median and the average of the 30 web thickness below the median. Calculate percentage by which each of the two averages differs from the average of all the web measurements (Wa).

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Method No. 505.1

COMPRESSIBILITY OF GRAINS

1. PROCEDURE

1.1 Select ten grains which are normal in shape, contain no obvious flaws, and are representative of the sample. Cut off each end of each grain so that the length of the central portion is equal to the diameter within 0.01 inch, and so that the ends are perpendicular to the sides within two degrees. In the course of preparation of the grains, do not subject the central portion, to be subsequently used for test, to sufficient pressure in any holding device such as a collet, chuck, vise, or pliers to leave a visible mark on this portion. The grain may be prepared for test by holding the end to be subsequently discarded firmly in a suitable holding device, cutting the grain with a hacksaw, and smoothing off each end by holding the central portion with the fingers while slowly rubbing the end against an abrasive surface, such as sandpaper. If the cutting is done with a lathe, by means of an abrasive or file, remove any burr remaining. CAUTION: Frequently test the temperature of any cutting tool or abrasive which is maintained in contact with the powder and allow it to cool whenever it becomes too warm to be held with comfort in contact with the back of the hand. Conduct the remainder of the test at a temperature within the limits of 60° to 100°F, making sure that the grains of powder come to the temperature of the room before testing if they have been stored at a different temperature. Measure the length of each grain to the nearest 0.001 inch by means of a micrometer caliper having jaws approximately 0.25 inch in diameter, holding the grain with the ends centered with respect to the jaws of the caliper. Compress each grain between parallel surfaces, increasing the load slowly until the first crack appears, as indicated by the appearance of a dark space between the walls of the crack when a beam of light is directed across the crack so that the direction of the beam is perpendicular to the place of the crack, and the line of vision is in the plane of the crack. (The probable location of the first crack can be determined by noting the first light colored area to form as pressure is increased). Release the pressure and immediately measure the length of the compressed grain. Calculate the compressibility of grain as follows:

$$\text{Compressibility of grain, percent} = \frac{100\ (A-B)}{A}$$

where

A = average length of uncompressed grains
B = average length of compressed grains

In case of doubt or dispute with respect to compliance of the sample for compressibility of grain, subject a total of 30 grains to the test at a temperature within the range of 70° to 80°F, and calculate the compressibility of grains of the 30 grains as directed above.

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Method No. 506.1

GRANULATION

1. PROCEDURE

1.1 Superimpose the required number of U. S. Standard sieves of whatever meshes required by specification and attach a receiving pan to the bottom layer. Place a 50 gram portion of the sample on the top sieve, cover this, and shake mechanically, or by hand, for five minutes. Weigh the material retained on each sieve and calculate the percentages of the sample passing through each sieve and, if required, that retained on any sieve. The mechanical shaker is geared to produce 300 $\pm$ 15 gyrations and 150 $\pm$ 10 taps of the striker per minute.

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Method No. 507.1

DENSITY OF LOADING

1. PROCEDURE

1.1 Determine volume of the cartridge case with water at 70° plus or minus 2°F. Load case with the powder under test to the same volume as for water capacity, using a procedure of pouring the powder into the case from a 2 inch height. If a certain charge is in use for a stated velocity then use the charge weight. Divide the powder unit by the volume weight of water and report as density.

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Method No. 508.1

AIR SPACE

1. PROCEDURE

1.1 This is free space in a cartridge after a weighed charge of powder is poured into the case from a 2 inch height and bullet inserted to a mean depth. Best taken by use of a gage, Drawing FB22274 - Gage, Air Space.

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GRINDING POWDER

1. PROCEDURE

1.1 Grind the powder in a mill* as follows: Insert a delivery tube with a 20 mesh sieve top, but do not attach the metal screw top of the jar. In this way, the mill is vented at both top and bottom. Place a glass plate in position and firmly clamp by the adjustable arm. Place a shield in the proper position. With the mill running at medium speed (belt on middle groove of pulley) introduce, into the hopper, a few grains of small size powder and allow a few seconds for them to grind before adding more grains of powder. The number of grains added at a time should be in proportion to size. This is determined by experience. Three to four samples can be ground before the cutter becomes warm. Allow sufficient time for the cutter to cool before grinding additional samples.

*When the Wiley mill is used for grinding powder, the intermediate size equipped with an explosion proof motor and shield is preferred.

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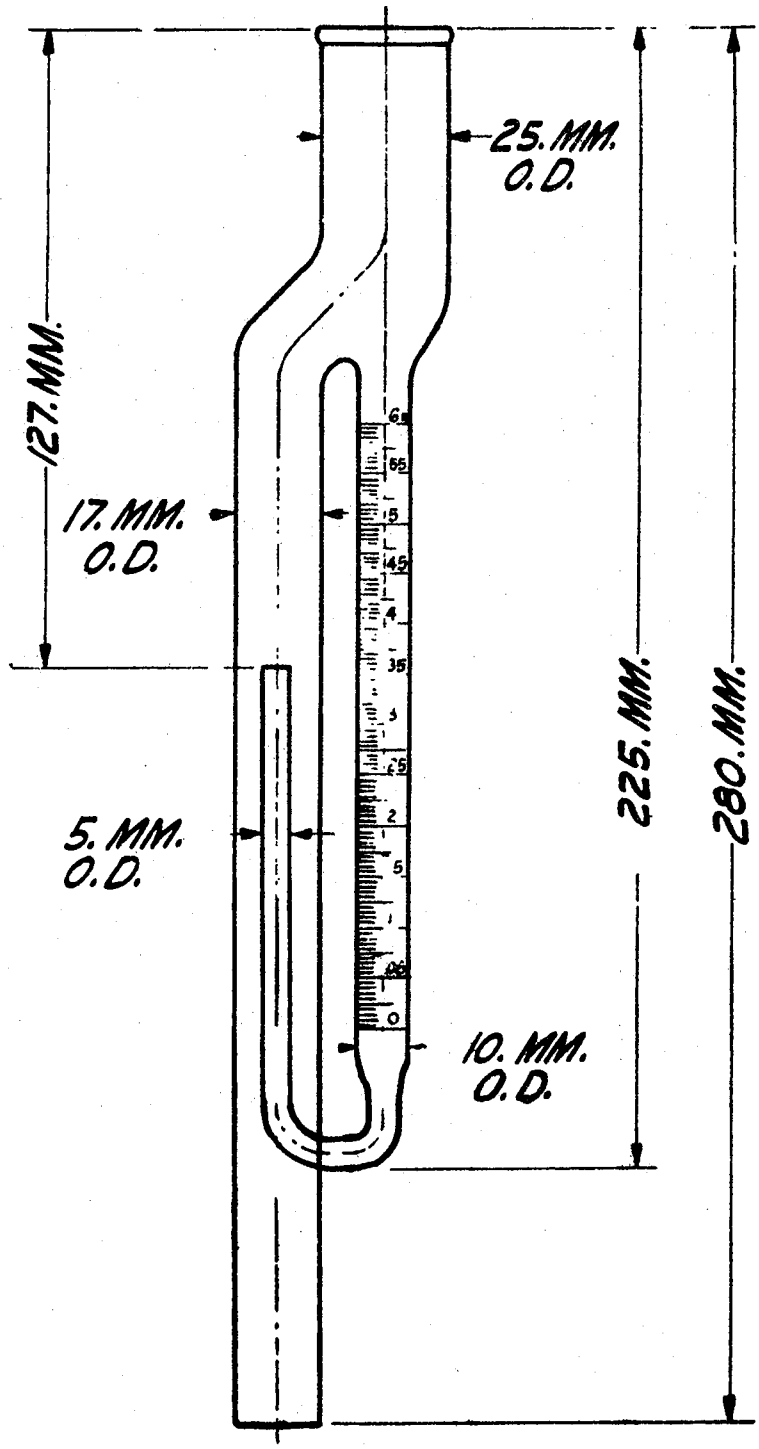


FIGURE 1
MOISTURE TUBE

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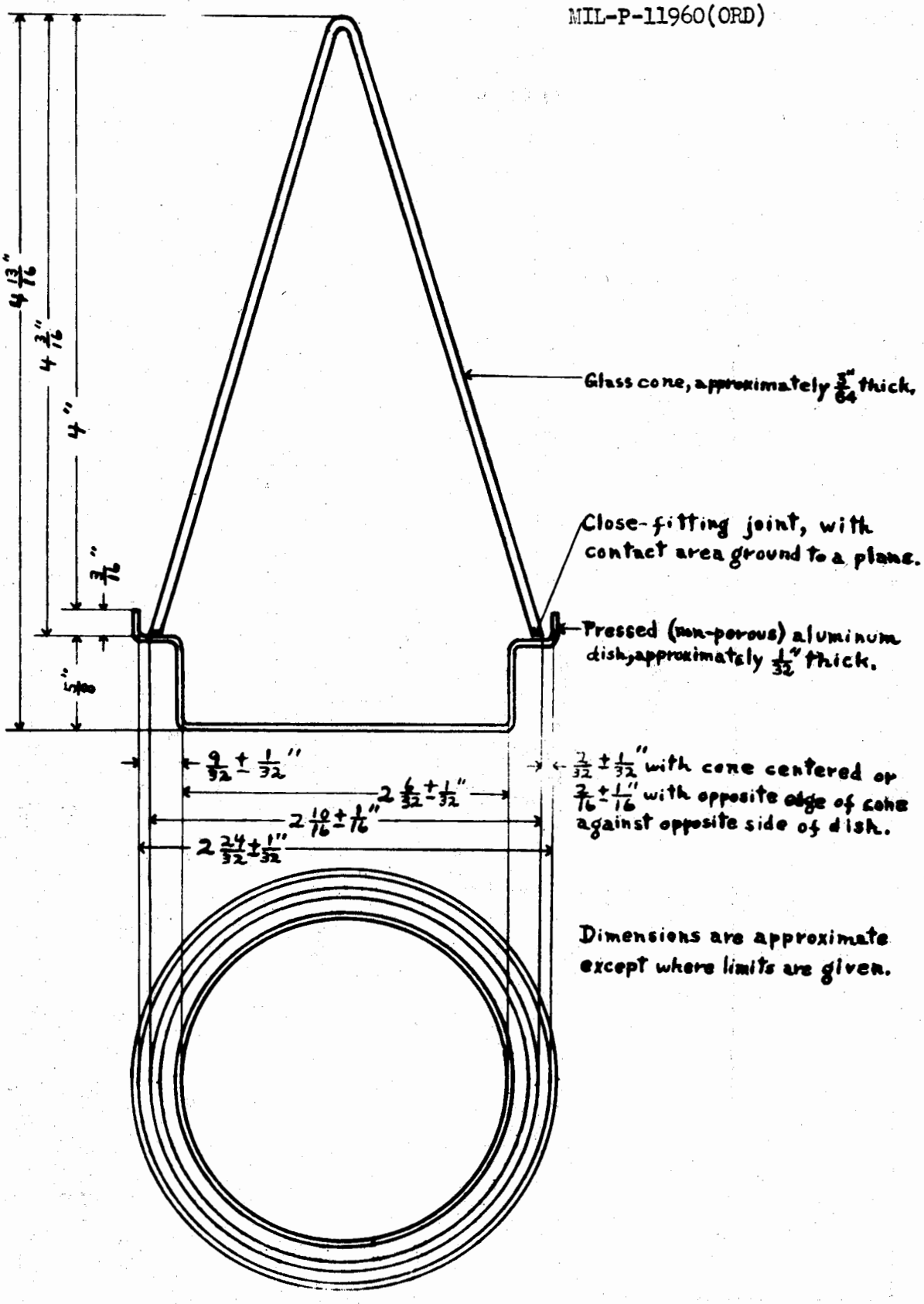


Figure 2. - Glass cone and aluminum dish for total volatiles determination.

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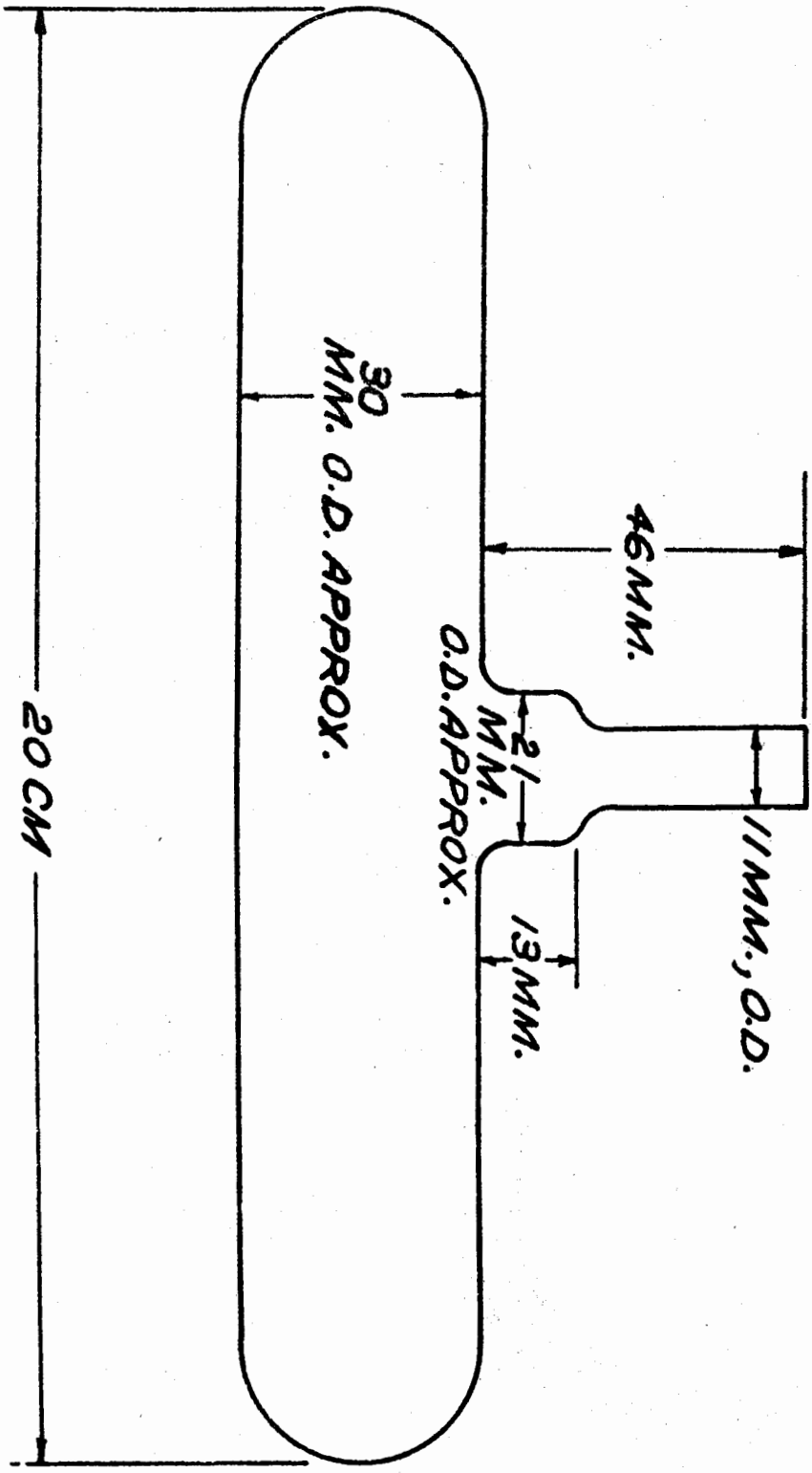


FIGURE 3

SOLUTION TUBE

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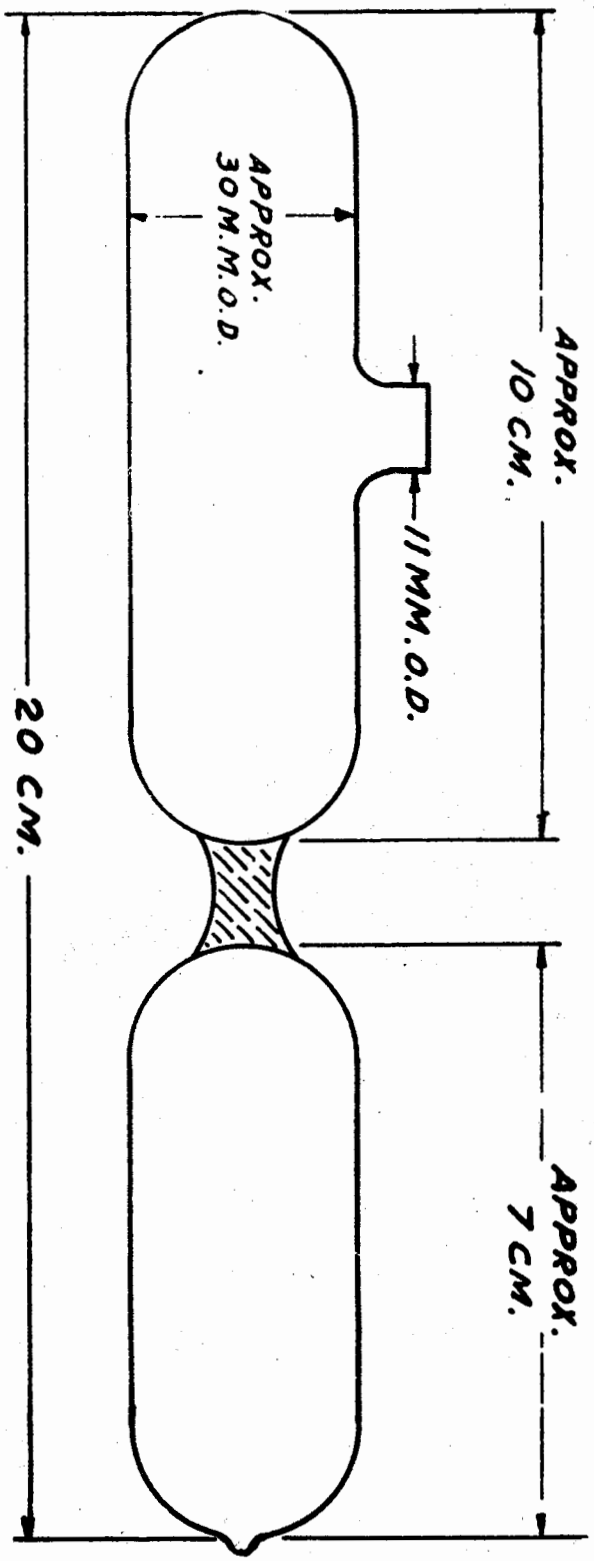


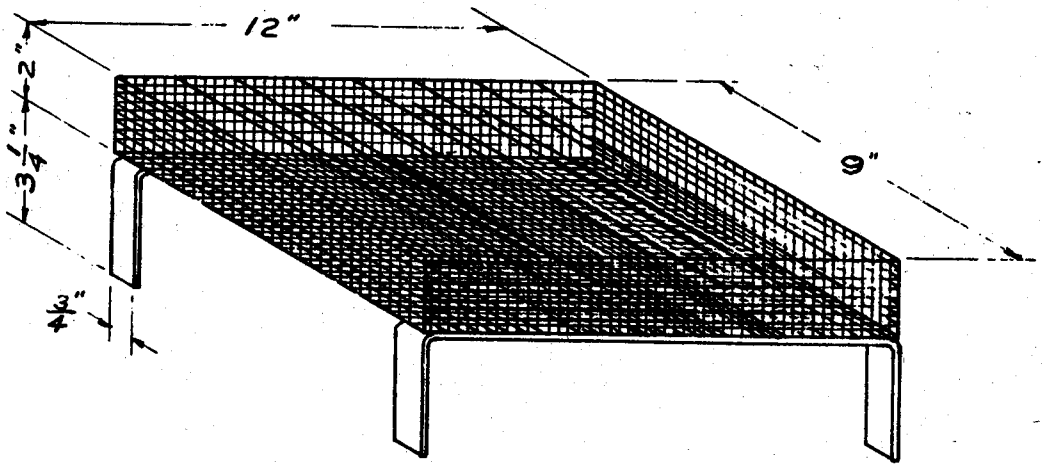
FIGURE 4

COUNTERPOISE

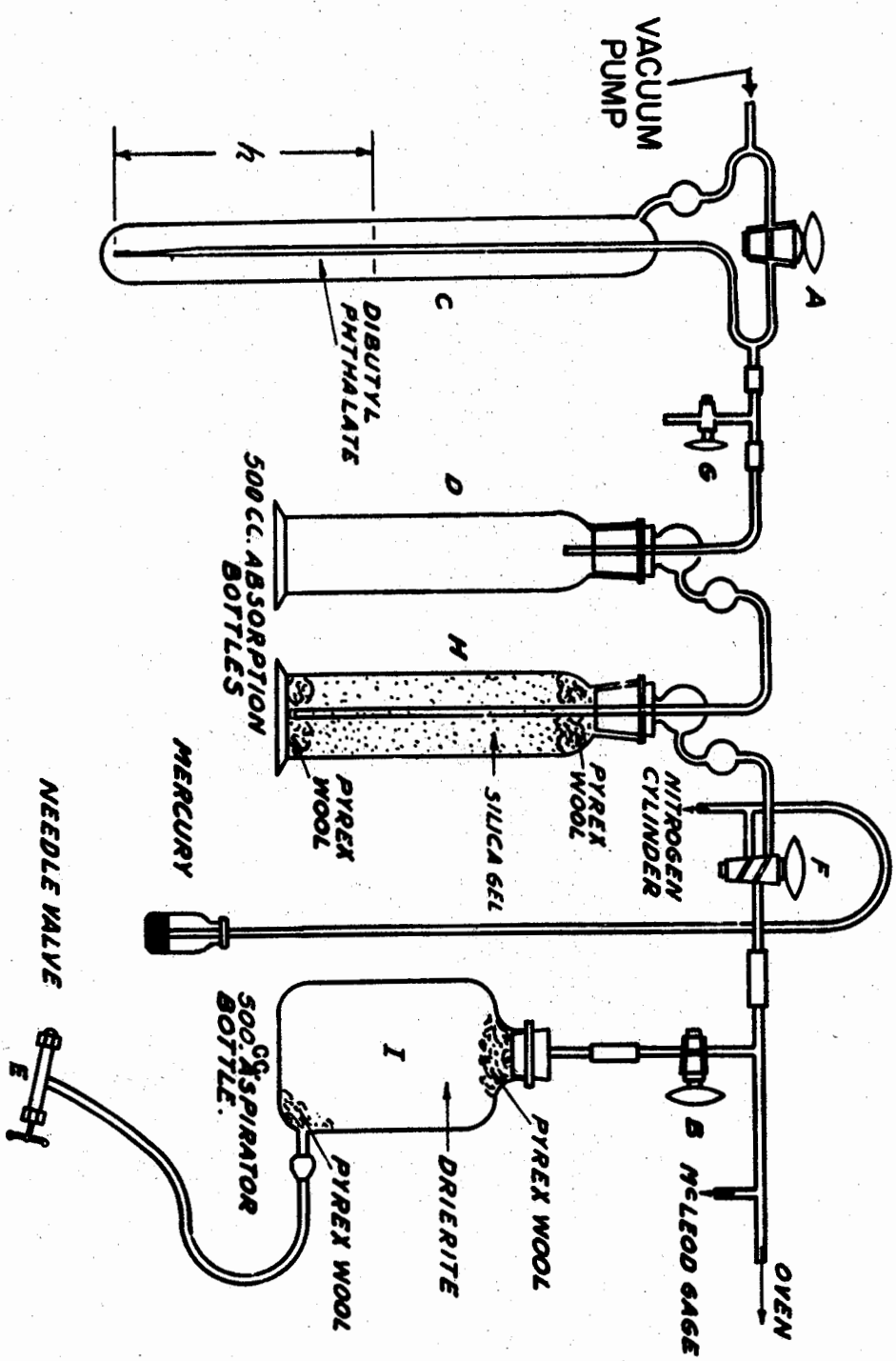
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WIRE SCREEN TRAY.

FIGURE 5



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VACUUM LINE ASSEMBLY.
FIGURE 6

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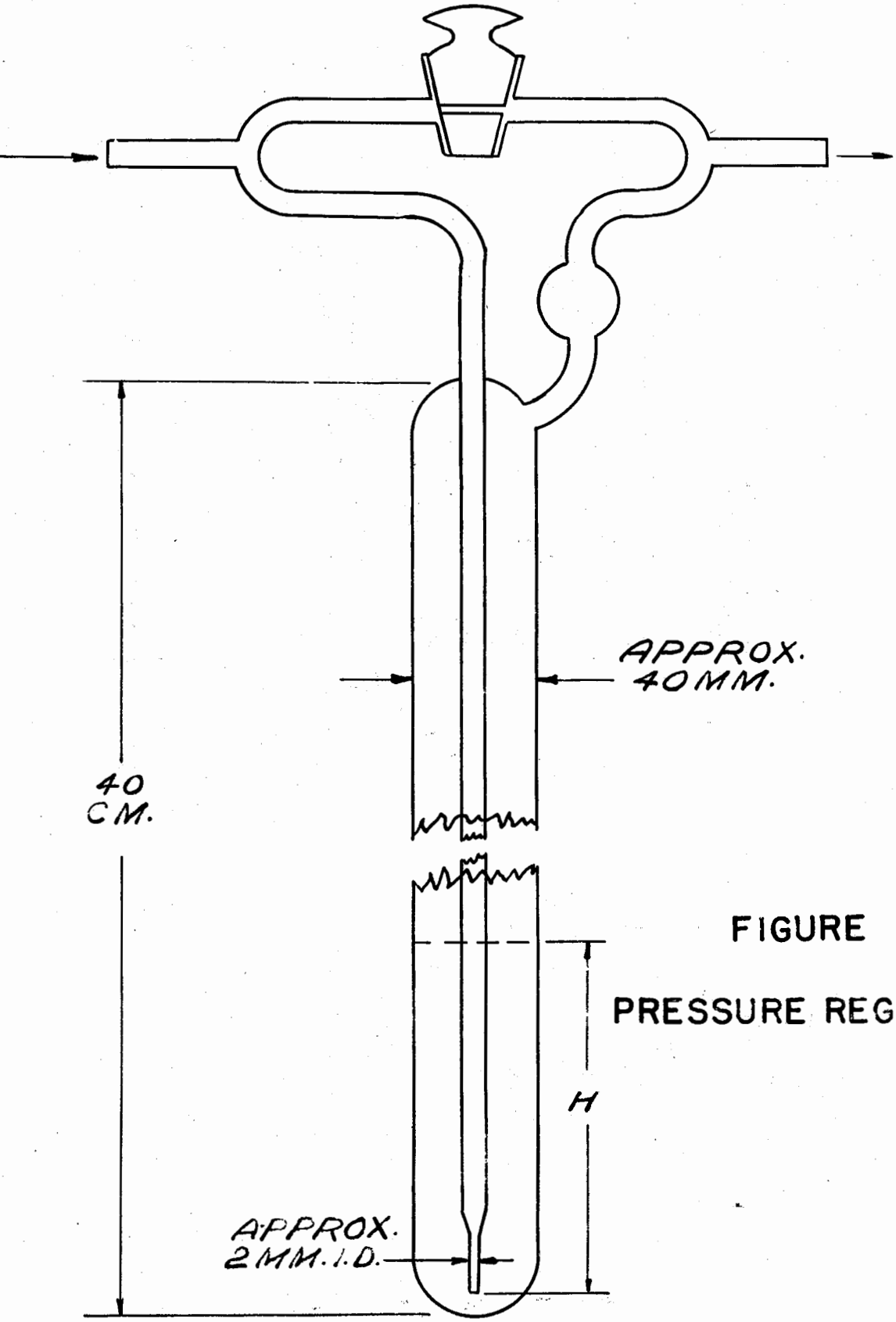
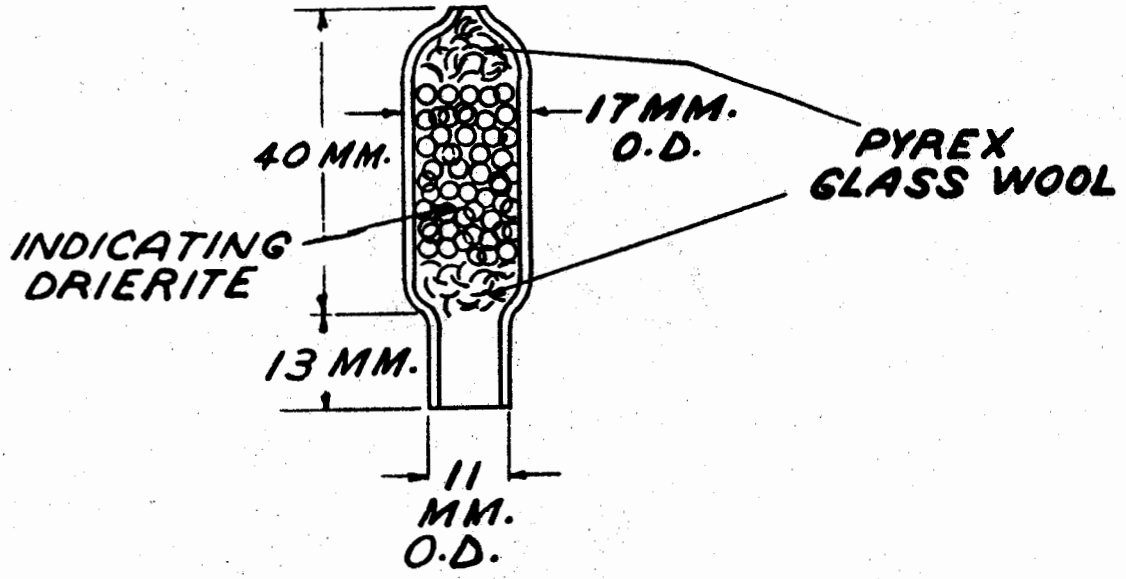


FIGURE 7

PRESSURE REGULATOR

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SMALL PROTECTIVE DRYING TUBE

FIGURE 8

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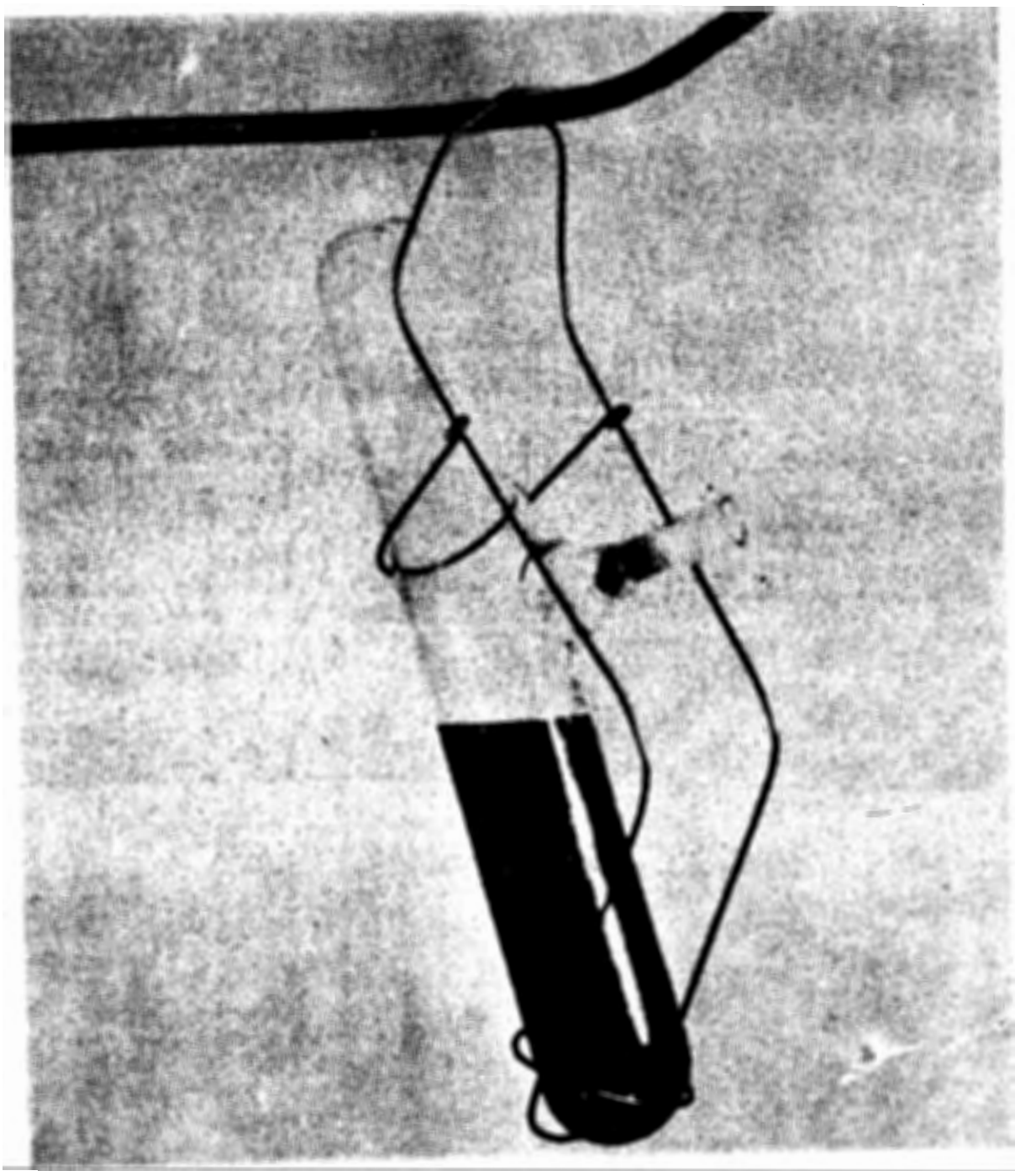


Figure 9. - Wire tube-holder.

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DRAWING 1.

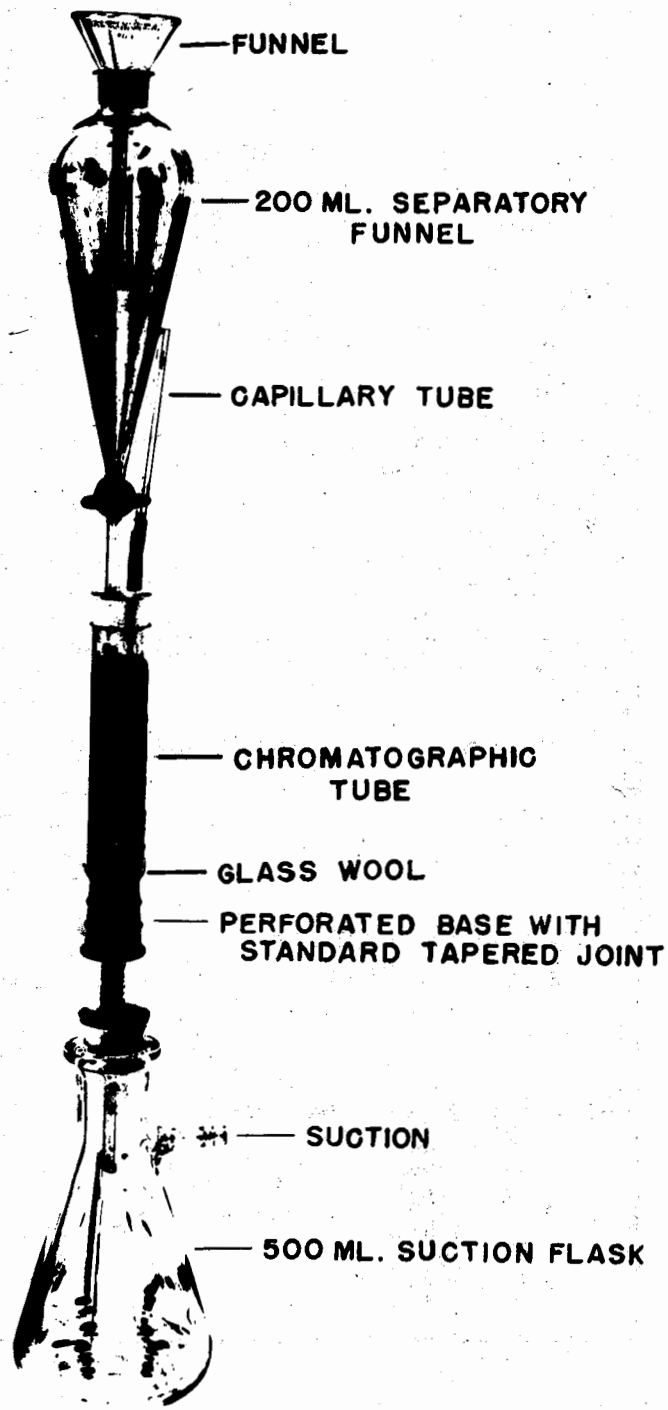
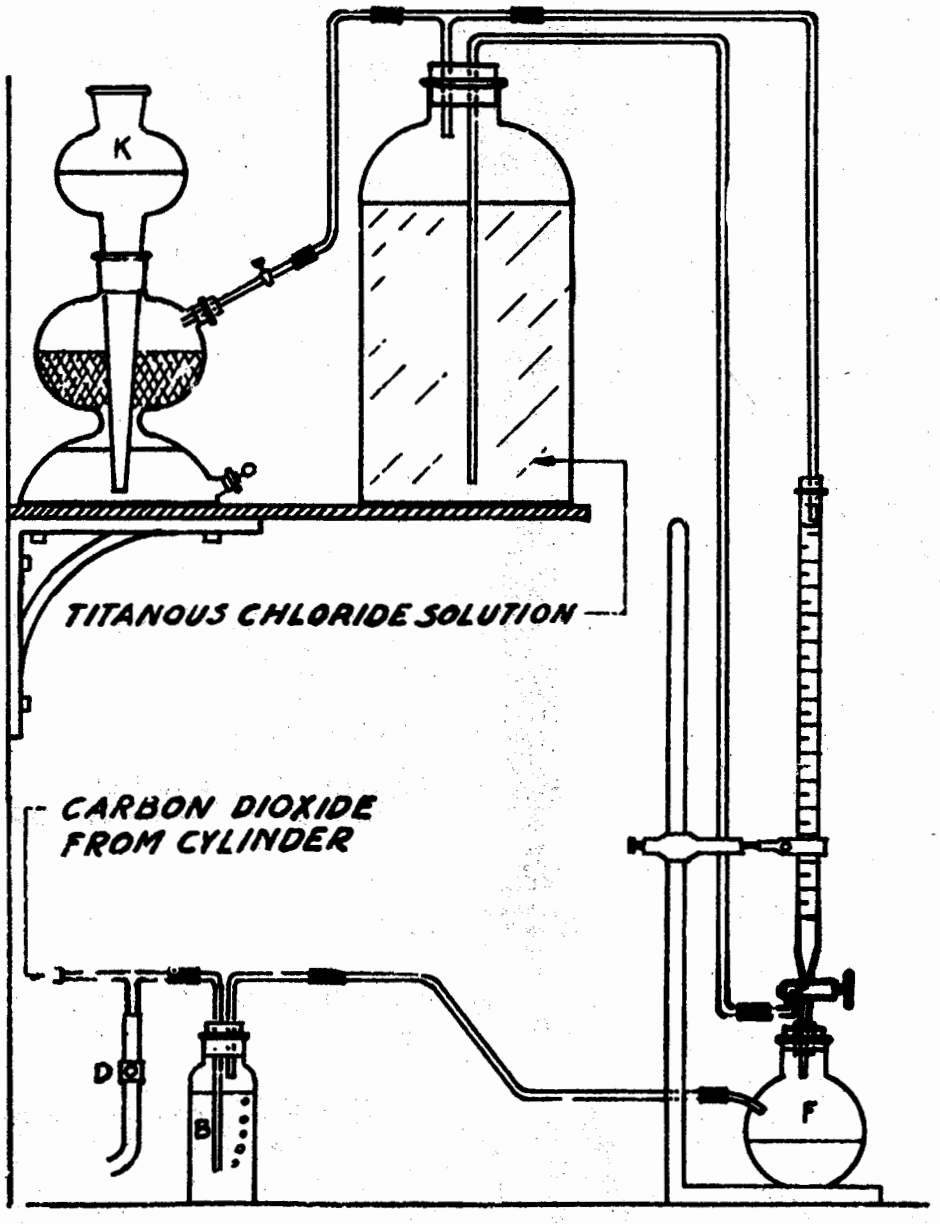


Figure 10. - Chromatographic assembly.

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Figure 11.

APPARATUS FOR STORING AND USING TITANOUS CHLORIDE SOLUTION.



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TITRATION FLASK

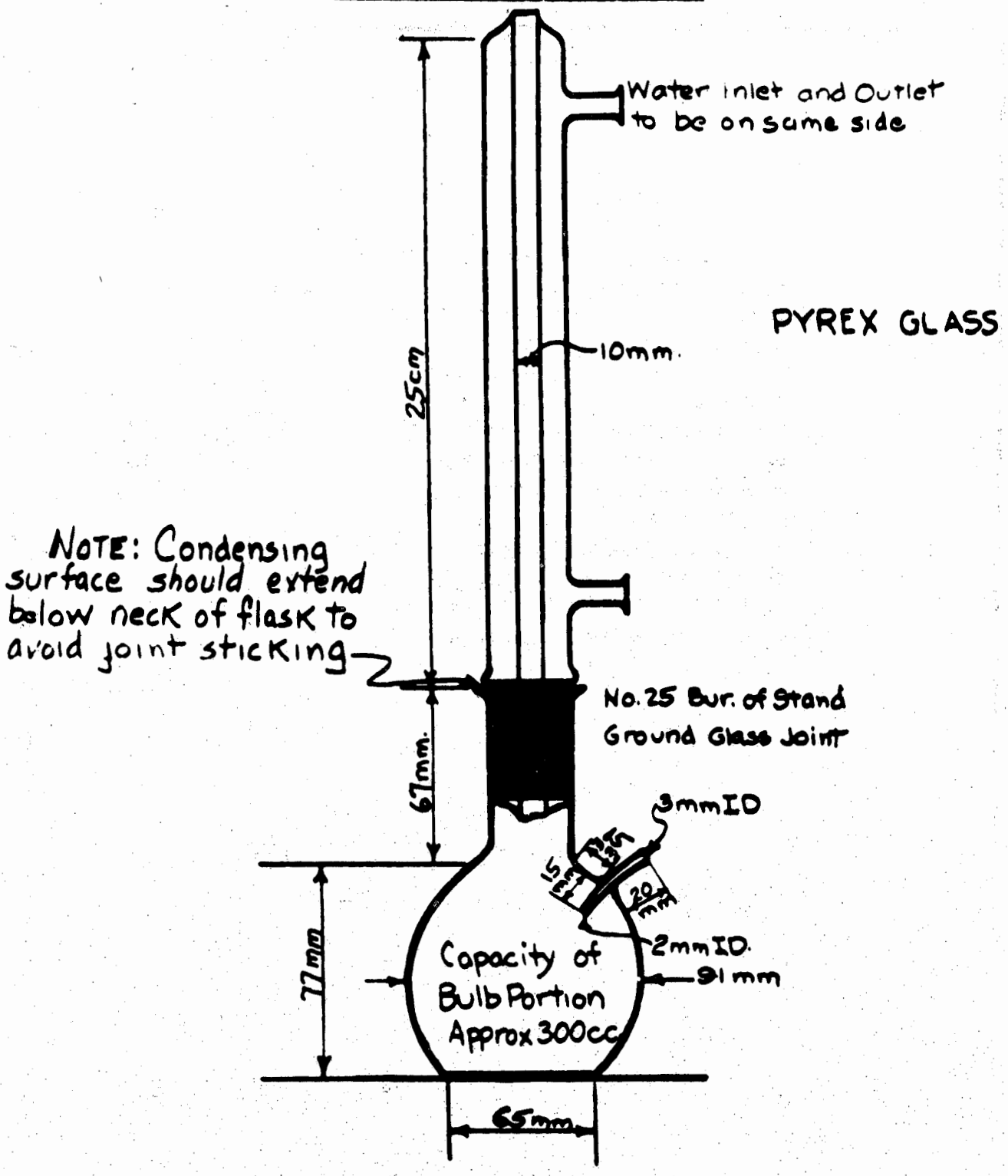
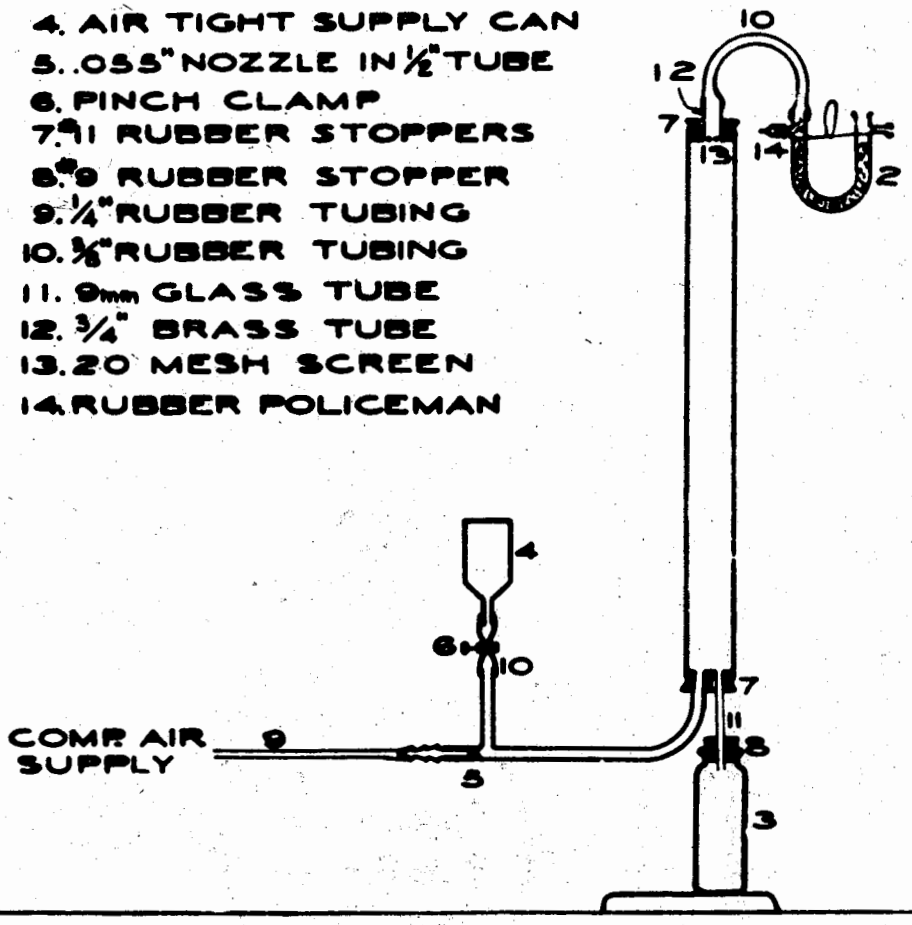


FIGURE 12

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LIST

1. COPPER TUBE, 2" O.D.
2. 6" U-TUBE
3. 16 OZ. GLASS BOTTLE
4. AIR TIGHT SUPPLY CAN
5. .055" NOZZLE IN ½" TUBE
6. PINCH CLAMP
7. #1 RUBBER STOPPERS
8. #9 RUBBER STOPPER
9. ¼" RUBBER TUBING
10. ⅜" RUBBER TUBING
11. 9mm GLASS TUBE
12. ¾" BRASS TUBE
13. 20 MESH SCREEN
14. RUBBER POLICEMAN



APPARATUS FOR THE DETERMINATION OF THE DUST CONTENT OF SMOKELESS RIFLE POWDERS

FIGURE 13

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DETERMINATION OF DUST IN SMALL ARMS SMOKELESS POWDER, WET METHOD.

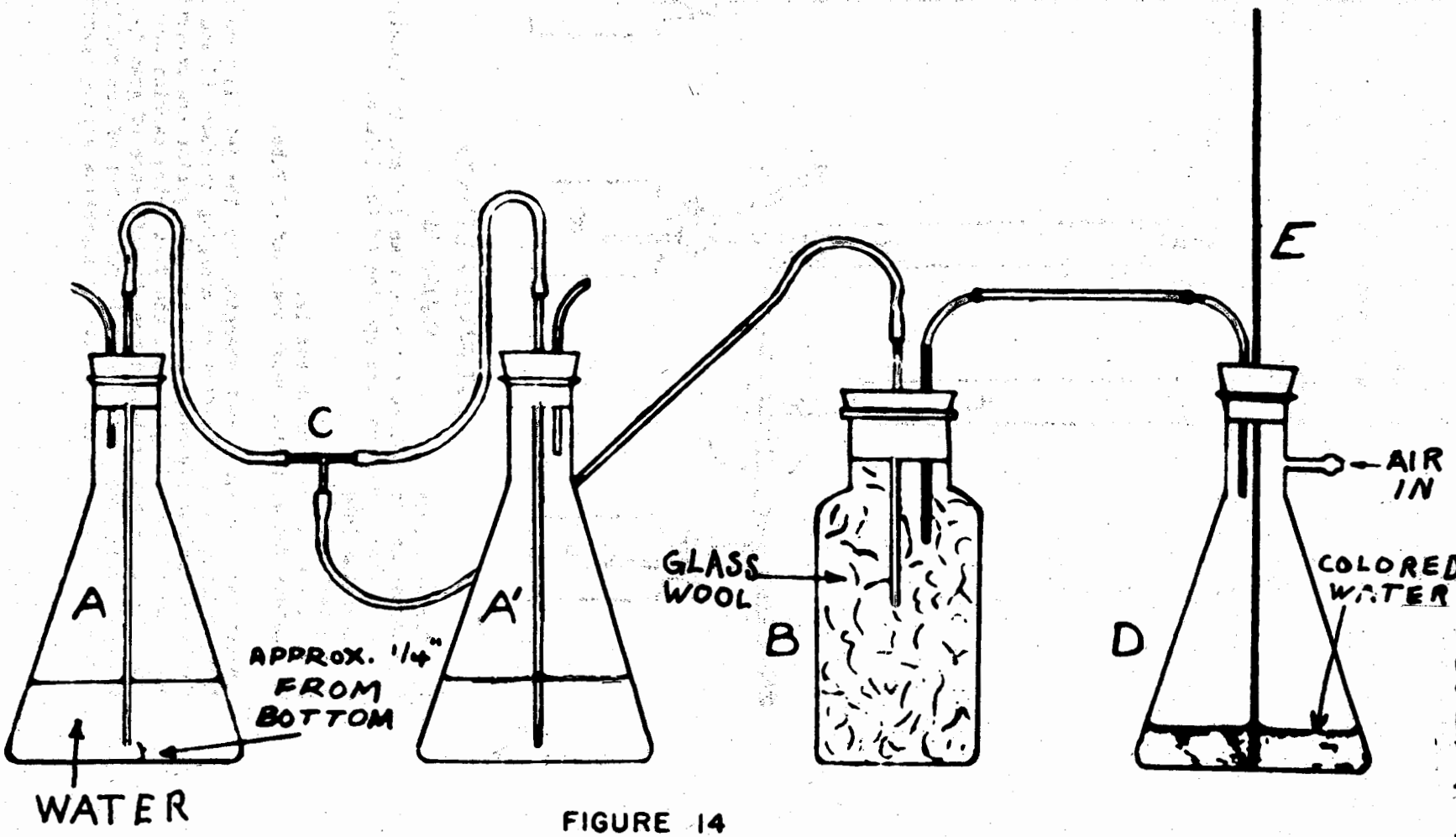
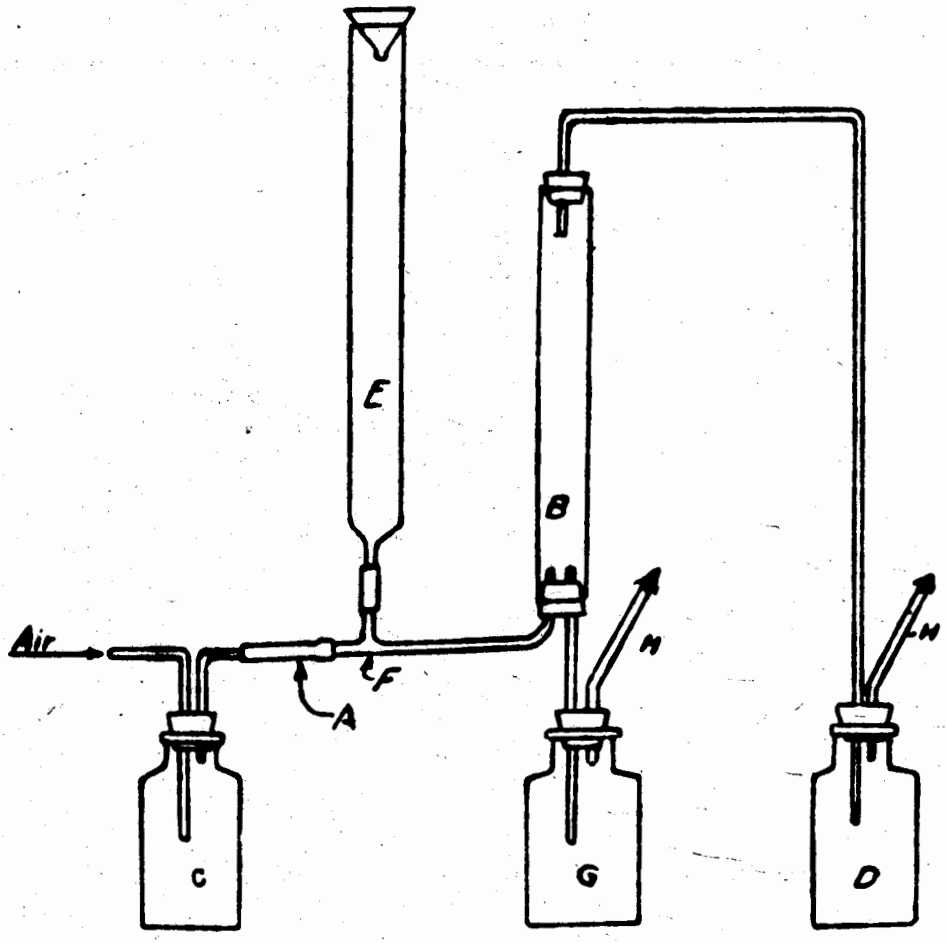


FIGURE 14

MIL-P-11960(ORD)

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DUST REMOVING APPARATUS

FIGURE 15

A. - Copper T-Tube.
F. - Copper nozzle, outlet not more than 1.5 mm. in diameter and placed not more than 2 mm. from the opening of the T-Tube.
E.- Charging Tube.
B. - Separating Tube.
C, D, G. - 8 oz. wide mouth bottle.
H. - Safety Tubes, cotton gauze over end of Tube.

Operations: The air stream emerging from the jet (A) produces a venturi effect, drawing the sample from the charging chamber (E) and expelling it into the large upright separator (B) where the grains are separated from the dust by virtue of their larger size, the dust being carried over into collector (D) while the whole grains, under the influence of gravity, fall back to collector (C). The cotton gauze is put on top of the safety tube (H) to insure that no grains can escape the apparatus during operation.

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